

INTRODUCTION TO BIOLOGY & THE LIVING WORLD — MEMORY TRICK SHEET

General Science · Biology · Chapter 1 · Sirf Yaad Rakhne Wali Cheezein — Tricks + Chains

Ye full notes nahi hai — source chapter ke coined-by vs father, branches, taxonomy, five kingdoms, Monera, symbiosis, virus–viroid–prion aur biological names ko exam-recall ke liye compress kiya gaya hai.

1. Biology Basics — Coined vs Father

- Biology = study of living organisms
- Greek roots: Bios = Life; Logos = Study
- Term “Biology” coined by Lamarck & Treviranus
- Father of Biology = Aristotle

TRICK - “Lamarck + Treviranus TERM; Aristotle FATHER”

Coined-by aur Father ko ek hi person mat samjho.

Bios + Logos = Life + Study.

EXAM TRAP -

Biology term = Lamarck & Treviranus; Father of Biology = Aristotle.

2. Three Core Branches — B-Z-M

- Botany → Plants → Theophrastus
- Zoology → Animals → Aristotle
- Microbiology → Microbes → Antonie van Leeuwenhoek

TRICK - “B-Z-M = Botany–Zoology–Microbiology”

Father chain: Theophrastus – Aristotle – Leeuwenhoek.

Visual hook: Plant → Animal → Microbe.

3. Sub-Branched — Person-to-Topic Anchors

Branch / Source Association	Study
Endocrinology — Thomas Addison	Hormones / Endocrine system
Immunology — Edward Jenner	Immunity
Genetics — Gregor Mendel	Heredity & variation
Cytology — Robert Hooke	Cells
Eugenics — Francis Galton	Human generations
Taxonomy — Carolus Linnaeus	Identify / Name / Classify
Paleontology — George Cuvier	Fossils
Embryology — Panchanan Maheshwari	Gametes, fertilisation, embryo
Agronomy — Pietro de Crescenzi	Crop production & soil
Agrostology — Johann Scheuchzer	Grasses
Anthology	Flowers
Serpentology — Patrick Russell	Snakes
Protozoology — A. van Leeuwenhoek	Protozoa

TRICK - “Addison–Jenner–Mendel–Hooke–Galton–Cuvier”

Hormone = Addison | Immunity = Jenner | Genetics = Mendel | Cell = Hooke | Eugenics = Galton | Fossil = Cuvier.

Grass = Agrostology; Snake = Serpentology; Protozoa = Protozoology.

4. Taxonomy — Coined vs Father + Binomial Rules

- Term “Taxonomy” coined by Augustin Pyramus de Candolle
- Father of Taxonomy = Carolus Linnaeus
- Taxonomy = Identifying + Naming + Classifying organisms
- Binomial Nomenclature introduced by Carolus Linnaeus; source references Systema Naturae
- Scientific name = Genus + Species
- Genus first letter capital; species small; scientific name italicized
- Example: Homo sapiens → Homo = Genus; sapiens = Species

TRICK - “de Candolle COINED; Linnaeus FATHER + BINOMIAL”

Taxonomy ka classic 3-way split: coined by de Candolle, father Linnaeus, binomial also Linnaeus.

EXAM TRAP -

Genus Capital + species small + whole scientific name italics.

5. Taxonomic Hierarchy — K-P-C-O-F-G-S

- Kingdom → Phylum (Animals) / Division (Plants) → Class → Order → Family → Genus → Species

TRICK - “King Philip Came Over For Good Soup”

K = Kingdom, P = Phylum, C = Class, O = Order, F = Family, G = Genus, S = Species.

EXAM TRAP - “Herbarium = Dry + Press”

Plants herbarium mein drying & pressing se preserve kiye jaate hain.

6. Five Kingdom Classification — Whittaker 1969

- Proposed by Robert Whittaker in 1969
- Criteria = Cell Structure + Body Organisation + Nutrition + Reproduction + Phylogenetic Relationship
- Five kingdoms = Monera, Protista, Fungi, Plantae, Animalia

TRICK - “C-B-N-R-P” Criteria

Cell → Body → Nutrition → Reproduction → Phylogeny.

TRICK - “M-P-F-P-A” Five Kingdoms

Monera → Protista → Fungi → Plantae → Animalia.

Feature	Monera	Protista	Fungi	Plantae	Animalia
Cell	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Wall	Peptidoglycan	Few present	Chitin	Cellulose	Absent
Nucleus membrane	Absent	Present	Present	Present	Present
Nutrition	Auto/Hetero	Auto/Hetero	Heterotroph	Autotroph	Heterotroph

TRICK - “Only MONERA = Prokaryotic”

Nuclear membrane absent only in Monera among the five kingdoms.

Wall material: Monera Peptidoglycan | Fungi Chitin | Plantae Cellulose | Animalia None.

EXAM TRAP -

Fungi wall = Chitin, NOT cellulose. Animalia = no cell wall.

7. Monera Groups — M-H-T / Cyanobacteria / Mycoplasma

- Methanogens → methane producing → Methanobacterium
- Halophiles → saline conditions → Halococcus
- Thermoacidophiles → hot sulphur springs → Sulfolobus
- Cyanobacteria → nitrogen fixing, blue-green autotrophic bacteria → Nostoc, Anabaena
- Nostoc & Anabaena fix atmospheric N₂ through Heterocysts
- Mycoplasma → no cell wall; parasitic or saprophytic

TRICK - “M-H-T = Methane–Salt–Hot Acid”

Methanogen = Methane; Halophile = Salt; Thermoacidophile = Hot acidic/sulphur.
Nostoc + Anabaena = Heterocyst = N₂ fixation.

EXAM TRAP -

Mycoplasma = no cell wall.

8. Symbiosis — Lichen vs Mycorrhiza

- Lichen = Fungi + Algae
- Fungi absorbs water/minerals; algae photosynthesizes
- Lichen = pioneer on bare rocks + SO₂ air-pollution indicator + litmus dye source
- Mycorrhiza = Fungi + roots of higher plants
- Fungi helps mineral absorption (especially phosphorus) and protection; plant provides shelter/nutrition
- Examples in source = Pinus and Orchids

TRICK - “Lichen = F + A; Mycorrhiza = F + Root”

Lichen extras: Rock pioneer + SO₂ indicator + Litmus.
Mycorrhiza extras: Phosphorus absorption + Pinus/Orchid.

9. Virus vs Viroid vs Prion — Composition First

Type	Discovery / Key Name	Composition	Disease / Note
Virus	Ivanowsky; TMV crystallized by Stanley	DNA or RNA + protein capsid	Obligate parasite; living/non-living link
Viroid	T.O. Diener	RNA only; no protein coat	Stunt disease in Chrysanthemum
Prion	Stanley B. Prusiner	Protein only; no nucleic acid	Scrapie; CJD; Kuru

TRICK - “V = Nucleic Acid + Coat | Viroid = RNA | Prion = Protein”

Composition se teenon ko pehle separate karo.
TMV: Ivanowsky = discovery; Stanley = crystallization.

EXAM TRAP -

Viroid = RNA only; Prion = protein only; Virus = nucleic acid + capsid.

10. Biological Names — Animal Memory Clusters

Common Name	Biological Name	Common Name	Biological Name
Peacock	Pavo cristatus	Frog	Rana tigrina
Cat	Felis catus	Dog	Canis lupus familiaris
Cow	Bos taurus	Buffalo	Bubalus bubalis
Goat	Capra aegagrus hircus	Tiger	Panthera tigris
Lion	Panthera leo	Leopard	Panthera pardus
Bear	Ursus thibetanus	Rabbit	Oryctolagus cuniculus
Elephant	Elephas maximus indicus	Dolphin	Platanista gangetica

TRICK - “Panthera Trio = Tiger–Lion–Leopard”

Tiger = Panthera tigris | Lion = Panthera leo | Leopard = Panthera pardus.
Same genus Panthera; species changes.

11. Biological Names — Plant Memory Clusters

Common Name	Biological Name	Common Name	Biological Name
Carrot weed	Parthenium hysterophorus	Pea	Pisum sativum
Mustard	Brassica campestris	Fig	Ficus carica
Tomato	Solanum lycopersicum	Rice	Oryza sativa
Banyan	Ficus benghalensis	Sugarcane	Saccharum officinarum
Onion	Allium cepa	Asafoetida (Hing)	Ferula asafoetida
Cotton	Gossypium	Coffee	Coffea arabica
Tea	Camellia sinensis	Turmeric (Haldi)	Curcuma longa

TRICK - “R-W-P-T = Oryza–Triticum–Solanum–Camellia”

Rice = Oryza sativa | Wheat = Triticum aestivum | Potato = Solanum tuberosum | Tea = Camellia sinensis.
Same-genus plant pair in listed names: Fig/Banyan = Ficus; Tomato/Potato = Solanum.

12. Five Taxonomic Example Anchors

Organism	Family	Order	Class	Phylum / Division
Human — Homo sapiens	Hominidae	Primates	Mammalia	Chordata
House Fly — Musca domestica	Muscidae	Diptera	Insecta	Arthropoda
Mango — Mangifera indica	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat — Triticum aestivum	Poaceae	Poales	Monocotyledonae	Angiospermae
Potato — Solanum tuberosum	Solanaceae	Solanales	Dicotyledonae	Angiospermae

TRICK - “Human–Fly–Mango–Wheat–Potato”

Animal examples: Human Chordata; House Fly Arthropoda.
Plant examples: Mango/Wheat/Potato = Angiospermae.

13. Final Exam Traps — 10 Red Flags

- Biology term: Lamarck & Treviranus ≠ Father Aristotle
- Taxonomy term: de Candolle ≠ Father Linnaeus
- Genus Capital; species small; scientific name italics
- Herbarium = drying & pressing
- Fungi wall Chitin | Plants Cellulose | Animalia no wall | Monera Peptidoglycan
- Only prokaryotic kingdom = Monera
- Lichen = Fungi + Algae; SO₂ indicator
- Virus = capsid+nucleic acid | Viroid = RNA only | Prion = protein only
- TMV: Ivanowsky discovery vs Stanley crystallization
- Panthera genus: tigris / leo / pardus

EXAM TRAP - “Coined-vs-Father | Wall | RNA/Protein | Same Genus”

Chapter ke maximum MCQ traps inhi four families se bante hain.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

Biology Bios+Logos; Lamarck/Treviranus term, Aristotle father -> B-Z-M: Theophrastus-Aristotle-Leeuwenhoek -> Addison hormone, Jenner immunity, Mendel genetics, Hooke cell, Galton eugenics, Cuvier fossil -> Taxonomy de Candolle coined / Linnaeus father+binomial; KPCOFGS -> Whittaker1969: C-B-N-R-P, M-P-F-P-A; Monera only prokaryotic; walls Peptidoglycan/Chitin/Cellulose/None -> M-H-T + Heterocyst + Mycoplasma no wall -> Lichen F+A/SO₂; Mycorrhiza F+root/P -> Virus nucleic acid+capsid / Viroid RNA / Prion protein; Ivanowsky discovery vs Stanley crystal -> Panthera trio; Rice Oryza, Wheat Triticum, Potato Solanum, Tea Camellia.

TOP 5: Biology term vs father | Taxonomy coined vs father | Monera only prokaryotic | Fungi=Chitin | Viroid RNA vs Prion Protein.

PLANT KINGDOM — MEMORY TRICK SHEET

General Science · Biology · Chapter 2 · Sirf Yaad Rakhne Wali Cheezein — Tricks + Chains

Ye full notes nahi hai — source chapter ke plant features, Cryptogam/Phanerogam split, Algae, Bryophyta, Pteridophyta, Gymnosperm, Angiosperm aur Monocot-Dicot facts ko exam-recall ke liye compress kiya gaya hai.

1. Plant Basics — S-C-P

- Plants = eukaryotic + multicellular + sessile
- Nutrition mainly photosynthetic autotrophic
- Cell wall = Cellulosic
- Plastids present

TRICK - “S-C-P = Sessile - Cellulose - Plastids”

Source chapter ke 3 highlighted plant features: Sessile + Cellulosic wall + Plastids.

EXAM TRAP -

Source notes ke comparison mein animals mein ye 3 highlighted features absent hain.

2. Big Split — Cryptogam vs Phanerogam

Group	Seeds	Reproductive Organ	Main Groups
Cryptogams	Absent	Hidden / inconspicuous	Thallophyta, Bryophyta, Pteridophyta
Phanerogams	Present	Cone or flower	Gymnosperm, Angiosperm

TRICK - “Crypto = Hidden + No Seed; Phanero = Visible + Seed”

Cryptogams reproduce by spores.

Phanerogams ke seeds present hote hain.

TRICK - “T-B-P / G-A”

Cryptogam trio = Thallophyta-Bryophyta-Pteridophyta.

Phanerogam pair = Gymnosperm-Angiosperm.

EXAM TRAP -

Gymnosperm = naked seed; Angiosperm = ovary/fruit covered seed.

3. Five Plant Groups — Generation + Vascular Memory

Group	Seed	Vascular	True Root/Stem/Leaf	Dominant
Thallophyta	No	No	No	Gametophyte
Bryophyta	No	No	No; rhizoids	Gametophyte
Pteridophyta	No	Yes	Yes	Sporophyte
Gymnosperm	Naked	Yes	Yes	Sporophyte
Angiosperm	Covered	Yes	Yes	Sporophyte

TRICK - “T-B = Gameto; P-G-A = Sporo”

Thallophyta + Bryophyta = Gametophyte dominant.

Pteridophyta + Gymnosperm + Angiosperm = Sporophyte dominant.

TRICK - “Vascular starts at P”

Thallophyta/Bryophyta = no vascular tissue.

Pteridophyta se Xylem + Phloem present.

4. Algae Basics — Thallus, Pectin-Cellulose, Pyrenoid, Holdfast

- Thallophyta/Algae body = undifferentiated thallus
- Vascular + mechanical tissues absent
- Mostly aquatic; gametophyte dominant
- Green algae wall = Outer Pectin + Inner Cellulose
- Green algae starch bodies = Pyrenoids in chloroplast
- Laminaria attaches to rocks by Holdfast

TRICK - "P-C-P-H"

Pectin outer → Cellulose inner → Pyrenoid starch → Holdfast attachment.

5. Algal Products — A-C-A + C-S

Product / Role	Source
Algin	Brown algae
Carrageenan	Red algae
Agar	Gelidium + Gracilaria (Red)
Protein / Space Food	Chlorella + Spirulina
Food	Porphyra + Sargassum

TRICK - "Algin Brown; Carrageenan Red; Agar G-G Red"

Agar = Gelidium + Gracilaria.

Protein/space food = Chlorella + Spirulina.

EXAM TRAP -

Algin aur Carrageenan dono hydrocolloids hain, but source algae color different hai.

6. Green-Brown-Red Algae — Pigment + Food

Type	Key Pigment	Stored Food	Examples
Green	Chl-a,b	Starch	Chlamydomonas, Volvox, Spirogyra
Brown	Fucoxanthin + Chl-a,c	Mannitol + Laminarin	Fucus, Ectocarpus, Sargassum, Laminaria
Red	Phycoerythrin + Chl-a,d	Floridean starch	Polysiphonia, Porphyra, Gracilaria

TRICK - "Brown = FUCO + MANNI-LAMI"

Brown algae unique pigment = Fucoxanthin.

Stored food = Mannitol + Laminarin.

TRICK - "Red = PHYCO + FLORIDEAN"

Red algae unique pigment = Phycoerythrin.

Stored food = Floridean starch.

EXAM TRAP -

Spirogyra = freshwater filamentous GREEN algae.

7. Bryophyta — Amphibian + Rhizoid + Gametophyte

- Wet/moist habitat
- True root, stem, leaves absent; rhizoids present
- Vascular tissue absent
- Gametophyte dominant; sporophyte depends on gametophyte
- Called Amphibians of Plant Kingdom
- Sexual reproduction = Oogamous

TRICK - “B = Bheega + Bina Vascular + Big Gametophyte”

Bryophyta ko wet habitat, absent vascular tissue aur gametophyte dominance se lock karo.

Type	Examples / Memory
Liverworts	Marchantia, Riccia
Mosses	Funaria = Cord Moss; Sphagnum = Bog Moss

TRICK - “Marchantia Liver; Funaria Cord; Sphagnum Bog”

Source notes Sphagnum ko peat moss/fuel history se bhi link karte hain.

EXAM TRAP -

Bryophyta vascular absent; Pteridophyta vascular present.

8. Pteridophyta — Vascular Cryptogam

- Source notes call them “Botanical Snakes of Plant Kingdom (Completely Terrestrial)”
- First land plants with vascular tissue
- Damp and shady places
- True root, stem, leaves present
- Xylem + Phloem present
- Seeds absent
- Water necessary for sexual reproduction
- Sporophyte dominant
- Male gametes formed in Antheridium
- Examples = Selaginella, Pteris, Equisetum, Psilotum

TRICK - “Pteri = Plant body Proper + Pipes Present, but Seed absent”

Proper root/stem/leaf + vascular pipes = Pteridophyta.

Still cryptogam, so seeds absent.

EXAM TRAP - “Equisetum = Only Living Sphenopsida”

Source marks this as a direct MCQ fact.

9. Gymnosperm — Naked Seed Pack

- Gymno = naked; Sperma = seed
- Seeds naked; ovary, fruit and true flower absent
- Sporophyte dominant
- Root, stem and needle-like leaves present
- Reproductive organ = unisexual cone
- Endosperm = Haploid (n)
- Companion cells of phloem and xylem vessels absent
- Evergreen woody trees/shrubs
- Examples = Cycas, Pinus, Ginkgo, Gnetum

TRICK - “GYM = Naked - No Ovary - No Fruit - Cone - n Endosperm”

Fruit absent because ovary absent.

EXAM TRAP -

Gymnosperm endosperm = Haploid (n), Angiosperm = Triploid (3n).

10. Angiosperm — D-T-3n

- Well-developed flowers + covered seeds
- Fruit/ovary present; seeds/ovules inside
- Double fertilisation present
- Triple Fusion present
- Endosperm = Triploid (3n)
- Divided into Monocot and Dicot

TRICK - “D-T-3 = Double fertilisation - Triple fusion - 3n”

Source chapter ki highlighted Angiosperm uniqueness.

11. Gymnosperm vs Angiosperm — Fast Contrast

Feature	Gymnosperm	Angiosperm
Seed	Naked	Covered
Flower	Absent	Present
Fruit	Absent	Present
Endosperm	Haploid (n)	Triploid (3n)
Double fertilisation	Absent	Present
Companion cells	Absent	Present

TRICK - “Gym = No Flower/Fruit/Double; Angio = All Present”

Endosperm contrast: n vs 3n.

12. Monocot vs Dicot — 1-3-P-F vs 2-4/5-R-T

Feature	Monocot	Dicot
Cotyledon	1	2
Root	Fibrous / Adventitious	Tap root
Vascular bundles	Scattered	Ring
Leaf venation	Parallel	Reticulate
Flower	Multiples of 3	Multiples of 4 or 5

TRICK - “MONO = 1 - 3x - Parallel - Fibrous”

One cotyledon, flower multiples of 3, parallel venation, fibrous roots.

TRICK - “DI = 2 - 4/5x - Reticulate - Tap”

Two cotyledons, flower multiples of 4/5, reticulate venation, tap root.

EXAM TRAP -

Monocot vascular bundles scattered; Dicot arranged in ring.

13. Crop Examples — Monocot vs Dicot

Monocot Examples	Dicot Examples
Corn / Maize	Rose
Wheat	Chickpea / Chana
Pearl Millet / Bajra	Tomato
Rice, Bamboo, Sugarcane	Mustard, Pea, Bean
Banana, Onion	Mango, Apple, Potato

TRICK - “W-C-R = Wheat-Corn-Rice = Monocot”

Rose-Tomato-Mustard = Dicot high-yield examples.

14. Final Exam Traps — 10 Red Flags

- Cryptogam = seed absent; Phanerogam = seed present
- Gymnosperm = naked seed/no ovary-fruit; Angiosperm = covered seed
- Thallophyta + Bryophyta = Gametophyte dominant; Pteridophyta + seed plants = Sporophyte dominant
- Algin Brown; Carrageenan Red; Agar Gelidium + Gracilaria
- Fucoxanthin Brown; Phycoerythrin Red
- Bryophyta vascular absent; Pteridophyta vascular present
- Equisetum = only living representative of Sphenopsida
- Gymnosperm endosperm n; Angiosperm endosperm 3n
- Angiosperm = Double fertilisation + Triple fusion + 3n endosperm
- Monocot = 1/Fibrous/Parallel/3; Dicot = 2/Tap/Reticulate/4-or-5

EXAM TRAP - “Seed - Generation - Pigment - Vascular - Endosperm - Cotyledon”

Chapter ke maximum confusion points in 6 families se bante hain.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Plants SCP = Sessile-Cellulose-Plastids → Crypto T-B-P no seed vs Phanero G-A seed → T-B Gametophyte, P-G-A Sporophyte; vascular starts at P → Algae P-C-P-H = Pectin/Cellulose-Pyrenoid-Holdfast; products Algin Brown, Carrageenan Red, Agar G-G, Chlorella-Spirulina protein → Green Starch; Brown Fucoxanthin + Mannitol/Laminarin; Red Phycoerythrin + Floridean → Bryophyta amphibian, rhizoid, no vascular, gametophyte; Marchantia liver, Funaria cord, Sphagnum bog → Pteridophyta first vascular land plant, true R-S-L, seed absent, sporophyte; Equisetum Sphenopsida → Gymnosperm naked/no ovary-fruit/cone/n endosperm → Angiosperm covered/flower-fruit/D-T-3n → Monocot 1-3-Parallel-Fibrous-Scattered; Dicot 2-4/5-Reticulate-Tap-Ring; Wheat-Corn-Rice mono, Rose-Tomato-Mustard dicot.”

TOP 5: Crypto vs Phanero | T-B Gameto vs P-G-A Sporo | Brown Fucoxanthin vs Red Phycoerythrin | Gym n vs Angio 3n | Mono 1/3/Parallel vs Dicot 2/4-5/Reticulate.

ANIMAL WORLD — MEMORY TRICK SHEET

General Science · Biology · Chapter 3 · Sirf Yaad Rakhne Wali Cheezein — Tricks + Chains

Ye full notes nahi hai — source chapter ke Non-Chordata, Chordata, Vertebrata, fishes, tetrapods, excretory organs, examples/scientific names aur class-wise comparison ko exam recall ke liye compress kiya gaya hai.

1. Animal Basics — E-M-H-M

- Animals = Eukaryotic + Multicellular + Heterotrophic + generally Motile
- Cell wall absent
- Plastids absent

TRICK - “E-M-H-M = Eukaryotic - Multicellular - Heterotrophic - Motile”

Plant contrast from source: animal cell mein cellulosic wall aur plastids absent.

2. Big Split — Notochord Absent vs Present

Division	Core Cue	Major Groups
Non-Chordata	Notochord absent	Porifera to Hemichordata
Chordata	Notochord present at some life stage	Urochordata, Cephalochordata, Vertebrata

TRICK - “NO Noto = Non-Chordata; Noto = Chordata”

Chordata ke hallmark 3 = dorsal notochord + dorsal hollow nerve cord + pharyngeal gill slits.

TRICK - “Herdmania + Amphioxus = Protochordates”

Herdmania = Urochordata; Amphioxus = Cephalochordata.

EXAM TRAP -

Hemichordata ko source notes connecting link between Non-chordates and Chordates batate hain.

3. Non-Chordata Order — 10 Phyla

- I Porifera
- II Cnidaria
- III Ctenophora
- IV Platyhelminthes
- V Aschelminthes / Nematoda
- VI Annelida
- VII Arthropoda
- VIII Mollusca
- IX Echinodermata
- X Hemichordata

TRICK - “Po-Cni-Cte-Pla-As-An-Ar-Mo-Ec-He”

10 phyla ko sound-chain mein bolo; order yaad rahega.

Size trap: Arthropoda largest, Mollusca second largest.

4. Porifera - Cnidaria - Ctenophora

Phylum	Standout Feature	Examples
Porifera	Choanocytes + canal system + high regeneration	Spongilla, Euspongia, Sycon
Cnidaria	Cnidocytes/Cnidoblasts; Polyp vs Medusa	Hydra, Physalia, Adamsia, Pennatula
Ctenophora	Bioluminescence + 8 comb rows	Ctenoplana

TRICK - “P-C-C = Collar - Cnidocyte - Comb”

Porifera = Collar cells.

Cnidaria = Cnidocytes.

Ctenophora = 8 Comb rows + light.

TRICK - “Polyp Sits, Medusa Floats”

Polyp = sessile; Medusa = free-floating.

EXAM TRAP -

Physalia = Portuguese man-of-war; Spongilla = freshwater sponge.

5. Flat - Round - Segmented Worms

Group	Body/Coelom	Excretion	Examples
Platyhelminthes	Flat; acoelomate	Flame cells	Taenia, Fasciola, Planaria
Aschelminthes	Round; pseudocoelomate	Renette cells	Ascaris, Wuchereria, Ancylostoma
Annelida	Segmented; true coelom	Nephridia	Nereis, Pheretima, Hirudinaria

TRICK - “Flat-Flame, Round-Renette, Ring-Nephridia”

Body form ko excretory organ ke saath pair karo.

TRICK - “Taenia Tapeworm; Wuchereria Filarial; Pheretima Earthworm”

High-yield example identity chain.

EXAM TRAP -

Hirudinaria = blood-sucking leech; source notes 33 segments highlight karte hain.

6. Excretion Super-Chain — F-R-N-M-B-P

Phylum	Excretory Structure
Platyhelminthes	Flame cells
Aschelminthes	Renette cells
Annelida	Nephridia
Arthropoda	Malpighian tubules
Mollusca	Bojanus organs
Hemichordata	Proboscis gland

EXAM TRAP - “F-R-N-M-B-P”

Flame → Renette → Nephridia → Malpighian → Bojanus → Proboscis.

Ye chapter ka sabse important mix-up chain hai.

7. Arthropoda — Largest Phylum

- Largest animal phylum; source notes: about two-thirds species are insects
- Chitinous exoskeleton + open circulatory system
- Body commonly head + thorax + abdomen
- Respiration via gills/book gills/book lungs/trachea
- Statocysts maintain body balance
- Excretion = Malpighian tubules

Example	Scientific Name / Note
Honey bee	Apis indica
Silkworm	Bombyx mori
Lac insect	Laccifera lacca
Mosquito vectors	Anopheles, Culex, Aedes
Living fossil	Limulus - King Crab
Other	Shrimp, Scorpion, Locusta

TRICK - “A-B-L = Apis-Bombyx-Laccifera”

Honey bee = Apis indica; Silkworm = Bombyx mori; Lac insect = Laccifera lacca.

EXAM TRAP -

Limulus = King Crab = living fossil.

8. Mollusca - Echinodermata - Hemichordata

Group	Memory Cue	Examples
Mollusca	2nd largest; CaCO ₃ shell; Ctenidia; Radula; Bojanus	Pila, Pinctada, Sepia, Octopus
Echinodermata	Water vascular system + tube feet; brain absent	Asterias, Echinus, Cucumaria, Ophiura
Hemichordata	Proboscis + Collar + Trunk; connecting link	Balanoglossus, Saccoglossus

TRICK - “Mollusc = C-R-B”

Ctenidia = respiration, Radula = grinding, Bojanus = excretion.

TRICK - “Echino = Water + Tube Feet + Aristotle”

Sea urchin food grinder = Aristotle’s lantern.

TRICK - “Hemi = P-C-T + Proboscis Gland”

Body = Proboscis-Collar-Trunk; excretion = Proboscis gland.

9. Vertebrata Tree — Jawless to Tetrapods

Branch	Group	Cue
Agnatha	Cyclostomata	Jawless; sucker-like circular mouth
Gnathostomata	Pisces + Tetrapoda	Jaws present
Pisces	Chondrichthyes + Osteichthyes	Fins
Tetrapoda	Amphibia + Reptilia + Aves + Mammalia	Four limbs / two pairs

TRICK - “A-G = Jawless-Jawed”

Agnatha = jawless; Gnathostomata = jawed.

TRICK - “Cyclo examples P-M”

Petromyzon = Lamprey; Myxine = Hag fish.

10. Fish — Chondrichthyes vs Osteichthyes

Feature	Chondrichthyes	Osteichthyes
Skeleton	Cartilage	Bone
Gill slits	5-7 pairs	4 pairs
Operculum	Absent	Present
Air bladder	Absent	Present
Heart	2 chambered	2 chambered
Temperature	Cold	Cold

TRICK - “Chondro = Cartilage + Cover NO + Air NO”

Operculum absent + air bladder absent = source ka direct trap.

TRICK - “Osteo = Bone + Both Present”

Operculum present + air bladder present.

Fish / Example	Special Identity
Torpedo	Electric organ
Trygon	Poison sting
Scoliodon	Dog fish
Pristis	Sawfish
Carcharodon	White shark
Exocoetus	Flying fish
Hippocampus	Sea horse
Labeo	Rohu
Clarias	Walking catfish
Betta splendens	Fighting fish

TRICK - “T-T = Torpedo Electric, Trygon Poison”

Exocoetus flies; Hippocampus horse; Clarias walks.

11. Tetrapoda Quick Comparison

Class	Heart	Temp.	Special
Amphibia	3	Cold	Water+land; mucous skin; tadpole
Reptilia	3 (Croc 4)	Cold	Dry keratin scales; CaCO ₃ egg
Aves	4	Warm	Feathers; hollow bones; toothless beak
Mammalia	4	Warm	Mammary glands + diaphragm; mostly viviparous

TRICK - “3-3-4-4 / Cold-Cold-Warm-Warm”

Amphibia-Reptilia = 3 chambered, cold.

Aves-Mammalia = 4 chambered, warm.

EXAM TRAP -

Reptiles generally 3-chambered, but Crocodile = 4-chambered.

12. Amphibia + Reptilia Examples

Class	Examples
Amphibia	Rana tigrina Frog; Bufo Toad; Hyla Tree frog; Ichthyophis Blindworm; Proteus
Reptilia	Naja naja Cobra; Typhlops Blind snake; Lizard; Chameleon; Tortoise; Dinosaur extinct reptile

TRICK - “R-B-H-I-P”

Amphibia: Rana-Bufo-Hyla-Ichthyophis-Proteus.

TRICK - “Naja = Cobra; Typhlops = Blind Snake”

Scientific/common-name direct pair.

13. Aves — EOPKC Flightless Chain

- 4-chambered heart + warm-blooded
- Dry skin with feathers
- Eggs with CaCO₃ covering
- Forelimbs = wings; jaws = toothless beak
- Hind limbs covered with scales
- Bones hollow/pneumatic

TRICK - “EOPKC = Emu - Ostrich - Penguin - Kiwi - Cassowary”

Source ka ready-made flightless birds mnemonic.

Ostrich = world's largest bird.

Bird	Scientific Name
Parrot	Psittacula
Peacock	Pavo cristatus
Pigeon	Columba

14. Mammalia — Mammary + Diaphragm + Monotreme

- 4-chambered heart + warm-blooded
- Hair/fur generally present
- Mammary glands = unique highlighted feature
- Diaphragm present
- Mostly viviparous
- Horns, hoofs, nails as exoskeletal structures

TRICK - “Mammal = Milk + Diaphragm + Mostly Live Birth”

Mammary gland aur diaphragm ko mammal signature pair banao.

EXAM TRAP - “P-E = Platypus-Echidna = Egg-laying Mammals”

Dono oviparous monotremes.

Mammal	Scientific Name
Human	Homo sapiens
Bat	Pteropus
Kangaroo	Macropus
Camel	Camelus

15. Scientific/Common Name Rapid Bank

Name	Remember As
Apis indica	Honey bee
Bombyx mori	Silkworm
Laccifera lacca	Lac insect
Pila	Apple snail
Pinctada	Pearl oyster
Asterias	Starfish
Echinus	Sea urchin
Petromyzon	Lamprey
Exocoetus	Flying fish
Hippocampus	Sea horse
Clarias	Walking catfish
Rana tigrina	Frog
Naja naja	Cobra
Pavo cristatus	Peacock
Homo sapiens	Human

TRICK - “Bee-Bombyx-Lac / Pila-Pearl / Asterias-Star / Naja-Cobra / Pavo-Peacock”

Name bank ko paired chunks mein revise karo.

16. Final Exam Traps — One-Glance

- Porifera = Choanocytes
- Ctenophora = Bioluminescence + 8 comb rows
- Platyhelminthes = Flame cells
- Arthropoda largest; Mollusca second largest
- Limulus = living fossil / King Crab
- Hemichordata = connecting link; Balanoglossus common example
- Chondrichthyes = no operculum + no air bladder; Osteichthyes = both present
- Crocodile = reptile with 4-chambered heart
- Flightless birds = EOPKC
- Platypus + Echidna = oviparous mammals

EXAM TRAP - “Cell - Light - Flame - Size - Fossil - Link - Fish - Croc - EOPKC - Monotreme”

Chapter ke 10 red-flag areas isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Animal E-M-H-M, wall/plastid absent → NonChordata no notochord vs Chordata N-N-G; Herdmania+Amphioxus protochordates → 10 phyla Po-Cni-Cte-Pla-As-An-Ar-Mo-Ec-He → P-C-C = Collar/Cnidocyte/Comb; Flat-Flame, Round-Renette, Ring-Nephridia → Excretion F-R-N-M-B-P → Arthropoda largest/chitin/open/Malpighian; Limulus fossil; Apis-Bombyx-Laccifera → Mollusc C-R-B, Echino water/tube feet/Aristotle, Hemi P-C-T/proboscis gland/link → Vertebrata: Agnatha Cyclo jawless vs Gnathostomata Pisces+Tetrapoda → Chondro cartilage/no operculum/no air vs Osteo bone/both present; Torpedo electric, Trygon poison, Exocoetus flying, Hippocampus sea horse, Clarias walking → Tetrapods 3-3-4-4 and Cold-Cold-Warm-Warm; Croc4 → Aves EOPKC + hollow bones → Mammal mammary+diaphragm; Platypus-Echidna oviparous.”

TOP 5: F-R-N-M-B-P excretion | Arthropoda largest vs Mollusca second | Chondro no/no vs Osteo yes/yes | Crocodile heart 4 | Platypus-Echidna oviparous mammals.

CELL — MEMORY TRICK SHEET

General Science · Biology · Chapter 4 · Sirf Yaad Rakhne Wali Cheezein — Tricks + Chains

Ye full notes nahi hai — source chapter ke cell discovery, size records, prokaryote/eukaryote, plant/animal comparison, membrane transport, organelles, plastids, cell cycle, mitosis-meiosis aur exam traps ko fast recall format mein compress kiya gaya hai.

1. Discovery Timeline — H-L-SV

Scientist	Year	Memory Fact
Robert Hooke	1665	Dead cork cells; coined Cellula
A. van Leeuwenhoek	1674	First living cells
Schleiden & Schwann	1838-39	Cell Theory
Rudolf Virchow	1855	New cells from pre-existing cells

TRICK - “Hooke Dead, Leeuwenhoek Live, S-S Theory, Virchow Old-to-New”

Order: 1665 > 1674 > 1838-39 > 1855.

Cellula = Hooke; Omnis cellula-e-cellula = Virchow.

EXAM TRAP -

Hooke = dead cork; Leeuwenhoek = first living cells.

2. Cell Records — O-M-N

- Largest cell = Ostrich egg; source note 15 x 13 cm, about 1.4 kg
- Smallest cell = Mycoplasma PLO; 0.1 micrometre
- Longest cell = Neuron; 90-100 cm
- RBC = Biconcave; about 7.2 micrometre

TRICK - “O-M-N = Ostrich - Mycoplasma - Neuron”

Largest > Smallest > Longest ko alag records ki tarah yaad karo.

EXAM TRAP -

Largest cell aur longest cell same nahi.

3. Prokaryote vs Eukaryote — 70/80 + Circle/Linear

Feature	Prokaryote	Eukaryote
Size	0.1-5 micrometre	10-100 micrometre
Nucleus	No true nucleus; nucleoid	True nucleus
Organelles	Membrane-bound absent	Present
Ribosome	70S	80S cytoplasm; 70S mito/plastid
DNA	One naked circular; no histone	Multiple linear; histone present
Respiration	Mesosoma - source note	Mitochondria

TRICK - “Pro = 70 + Circle + No Histone; Eu = 80 + Linear + Histone”

Eukaryote exception: mitochondria/plastids ke ribosomes 70S.

TRICK - “70 = 50+30; 80 = 60+40”

Ribosome subunit trap ko direct number pair se lock karo.

EXAM TRAP -

Eukaryotic cytoplasm 80S, but mitochondria/plastids 70S.

4. Plant vs Animal — W-P-V-C

Feature	Plant	Animal
Cell wall	Present - cellulose	Absent
Plastid	Present	Absent
Vacuole	Large central	Very small/absent
Centrosome	Absent	Present
Golgi	Smaller; Dictyosomes	Larger
Mitochondria	Fewer	More

TRICK - “Plant = Wall + Plastid + big Vacuole; Animal = Centrosome”

Plant Golgi = Dictyosomes.

EXAM TRAP -

Centrosome animal; plastid and cell wall plant.

5. Membrane Transport — D-O-E-E

Process	Direction / Requirement	Hook
Diffusion	High to Low; no energy	Gas > Liquid > Solid
Osmosis	Water across semipermeable membrane	Root water absorption
Endocytosis	Into cell; energy-dependent	Phago solid / Pino liquid
Exocytosis	Out of cell	Cell vomiting

TRICK - “D-O-E-E = Diffuse - Osmose - Enter - Exit”

Endocytosis enters; exocytosis exits.

TRICK - “Phago = Food/Solid; Pino = Pani/Liquid”

Dono Endocytosis ke types.

EXAM TRAP -

Phagocytosis solid; Pinocytosis liquid.

6. Osmosis — Hypo Swell, Iso Same, Hyper Shrink

Solution	Condition	Cell Effect
Hypotonic	Outside concentration < cell	Swells
Isotonic	Equal	No change
Hypertonic	Outside concentration > cell	Shrinks

TRICK - “HYPO = Hawa Bharo, HYPER = Hatao Pani”

Hypotonic swells; Hypertonic shrinks; Isotonic same.

- Aquaporins = specialised water-channel proteins responding to osmotic gradient

7. Cell Boundary + Protoplasm

- Cell wall = plant only; cellulose; permeable and protective
- Cell membrane = all cells; very thin, flexible, selectively permeable
- Source composition = Protein 50% + Lipid 40% + Carbohydrate 10%
- Protoplasm term coined by J.E. Purkinje, 1839; source notes discovery by Felix Dujardin
- Huxley called protoplasm the physical basis of life
- Cytoplasm = protoplasm excluding nucleus

- Nucleoplasm = protoplasm inside nucleus

TRICK - “50-40-10 = Protein-Lipid-Carb”

Cell membrane composition ka source-number pack.

TRICK - “Purkinje Protoplasm; Huxley Physical Basis”

Coiner aur quote ko pair mein yaad karo.

8. Nucleus — Brown 1831, DNA+Histone, 23 Pairs

- Nucleus discovered by Robert Brown, 1831
- Control centre / brain of cell
- Nuclear membrane = double membrane with pores
- Nucleolus discovered by Fontana; synthesises r-RNA
- Chromatin condenses into chromosomes during division
- Chromosome = DNA + Histone; Gene = functional DNA segment
- Human = 23 pairs = 22 autosome pairs + 1 sex-chromosome pair
- Male XY; Female XX
- Source notes: Safranin used for nucleus staining

TRICK - “23 = 22 + 1; Male XY, Female XX”

Human chromosome pack.

TRICK - “Brown Nucleus, Fontana Nucleolus”

Nucleus vs nucleolus discoverer trap.

EXAM TRAP -

Nucleus largest organelle, but large plant vacuole can be exception in source notes.

9. ER + Ribosome — Rough Protein, Smooth Lipid

- ER discovered by Keith R. Porter, Albert Claude & Ernest Fullam, 1945
- RER = ribosomes present; protein/enzyme synthesis
- SER = ribosomes absent; lipid/steroid synthesis + detoxification
- Ribosome discovered by G.E. Palade, 1953
- Ribosome = membrane-less; protein synthesis
- Polysome = chain of ribosomes attached to m-RNA

TRICK - “Rough = Ribosome = Protein; Smooth = Steroid/Lipid”

RER-P, SER-LS.

TRICK - “Palade 1953 = Protein Factory”

Ribosome discoverer + function pair.

10. Golgi — 1898 Post Office

- Discovered by Camillo Golgi, 1898
- Plant Golgi = Dictyosomes
- Cis face = convex, toward nucleus
- Trans face = concave, toward cell membrane
- Secretion, storage, packaging, processing
- Forms lysosomes and sperm acrosome
- Zone of exclusion = cytoplasm around Golgi with no organelles

TRICK - “Cis = Comes from nucleus; Trans = Travels to membrane”

Shape hook: Cis convex, Trans concave.

TRICK - “Golgi Makes L-A”

L = Lysosome; A = Acrosome.

EXAM TRAP -

Golgi = Post office / Traffic police of cell.

11. Lysosome + Mitochondria — Suicide vs Power

Organelle	Discoverer / Year	Core Memory
Lysosome	Christian de Duve, 1955	Suicidal bag; hydrolytic enzymes
Mitochondria	Albert von Kolliker, 1857	Powerhouse; ATP; Cristae

- Lysosome digests food, foreign particles, damaged organelles, bacteria, viruses and waste
- Mitochondria = double membrane; inner folds = Cristae
- ATP = Adenosine Triphosphate; energy currency
- Mitochondria semi-autonomous: own circular naked DNA, RNA and 70S ribosomes
- Source notes also link mitochondria with apoptosis

TRICK - “De Duve Suicide 1955; Kolliker Power 1857”

Lysosome = digestion/self-digestion; Mito = ATP.

EXAM TRAP -

Cristae = mitochondrial inner folds.

12. Plastids — C-C-L + A-E-A

Plastid	Pigment / Colour	Main Role
Chloroplast	Green; chlorophyll	Photosynthesis - Kitchen
Chromoplast	Colour except green	Fruit/flower colours
Leucoplast	Colourless	Storage

TRICK - “C-C-L = Chloroplast - Chromoplast - Leucoplast”

Green - Colour - Colourless.

TRICK - “A-E-A = Amyloplast - Elaioplast - Aleuroplast”

Amyloplast = starch; Elaioplast = oil; Aleuroplast = protein.

- Source notes credit plastid discovery to Ernst Haeckel and Schimper
- Plastids = double-membrane, semi-autonomous, plant cells only

EXAM TRAP -

Chloroplast = Kitchen; Mitochondria = Powerhouse; Lysosome = Suicidal bag; Golgi = Post office.

13. Vacuole + Centrosome + Peroxisome

- Vacuole membrane = Tonoplast
- Vacuole = storage house / dustbin
- Plant vacuole may occupy 60-90% cell volume; helps turgidity
- Animal vacuoles very small or absent
- Centrosome forms poles during animal-cell division

- Centrioles = 9 + 0 arrangement of triplet microtubules
- Peroxisomes in liver detoxify H_2O_2 to $H_2O + O_2$

TRICK - “Tono = Vacuole Top Cover”

Tonoplast = vacuole membrane.

TRICK - “9+0 = Centriole; H_2O_2 = Peroxisome”

Two direct one-liner facts.

14. Cell Cycle — G1-G0-S-G2-M

Phase	Memory Event
G1	Growth; comparatively longest after division
G0	Quiescent / non-dividing
S	DNA + centrosome replication; histone synthesis
G2	Prepare for mitosis; proteins/organelles; DNA check
M	Karyokinesis + Cytokinesis

TRICK - “Grow - Stop - Synthesize DNA - Get Ready - Multiply”

$G1 > G0 > S > G2 > M$.

- Plant-cell cytokinesis = Phragmoplast forms during new cell-wall construction

15. Mitosis vs Meiosis — $2n$ Same vs n Half

Feature	Mitosis	Meiosis
Cells	Somatic	Germ
Divisions	1	2
Chromosome	$2n$ to $2n$	$2n$ to n
Products	2 identical	4 haploid, genetically different
Purpose	Growth + repair	Gamete formation + variation

TRICK - “MITO = Maintain Number; MEIO = Make Half”

Mitosis keeps chromosome number same.

Meiosis halves it.

TRICK - “P-M-A-T”

Karyokinesis stages: Prophase > Metaphase > Anaphase > Telophase.

- Walther Flemming discovered mitosis by studying chromosomes
- Uncontrolled mitosis = Cancer
- Source notes link Turner and Klinefelter syndromes with meiotic chromosome-segregation errors
- Meiosis-I Prophase-I: Leptotene stage = chromosome condensation

EXAM TRAP -

Mitosis = 2 identical; Meiosis = 4 haploid/different.

16. Final Exam Traps — One-Glance

- Hooke dead cork vs Leeuwenhoek first living cells
- Largest Ostrich egg; smallest Mycoplasma PPLO; longest Neuron
- 70S prokaryote + mito/plastid; 80S eukaryotic cytoplasm
- Centrosome animal; plastid/cell wall plant
- Phagocytosis solid; Pinocytosis liquid
- Mito Powerhouse; Lysosome Suicidal; Golgi Post office; Chloroplast Kitchen

- Golgi forms lysosome + sperm acrosome
- Plant vacuole 60-90%; membrane Tonoplast
- Mitosis $2n$ to $2n$; Meiosis $2n$ to n
- Uncontrolled mitosis = Cancer

EXAM TRAP - “Dead-Live / O-M-N / 70-80 / Plant-Animal / Phago-Pino / Organelle Nicknames / Division”

Chapter ke major red flags isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Hooke1665 dead Cellula > Leeuwenhoek1674 living > Schleiden-Schwann1838-39 theory > Virchow1855 old-to-new → records O-M-N = Ostrich-Mycoplasma-Neuron → Pro 70/circular/no histone/mesosome vs Eu 80/linear/histone/mitochondria; mito-plastid 70S → Plant wall+plastid+big vacuole vs Animal centrosome → transport D-O-E-E; Hypo swell, Iso same, Hyper shrink; Phago solid/Pino liquid → membrane 50-40-10; Purkinje protoplasm/Huxley physical basis → Brown nucleus1831, Fontana nucleolus, humans 23=22+1 XY/XX → RER protein, SER lipid-steroid; Palade ribosome1953 → Golgi1898 Post Office, Cis nucleus/Trans membrane, forms Lysosome+Acrosome → De Duve lysosome1955 suicide vs Kolliker mito1857 powerhouse/Cristae/ATP/own DNA-RNA-70S → plastids C-C-L, leucoplast A-E-A; Tonoplast vacuole, centriole9+0, peroxisome H_2O_2 → cycle G1-G0-S-G2-M → Mitosis $2n-2n/2$ identical vs Meiosis $2n-n/4$ haploid; PMAT; cancer = uncontrolled mitosis.”

TOP 5: Hooke dead vs Leeuwenhoek living | 70S/80S exception | Phago solid vs Pino liquid | Golgi makes lysosome+acrosome | Mitosis same vs Meiosis half.

TISSUE — MEMORY TRICK SHEET

General Science · Biology · Chapter 5 · Sirf Yaad Rakhne Wali Cheezein — Tricks + Chains

Ye full notes nahi hai — source chapter ke Plant + Animal tissues, Meristematic/Permanent tissue, Xylem-Phloem, Epidermis-Cork, Epithelium, Connective tissue, Muscle aur Nervous tissue ke high-confusion facts ko fast recall format mein compress kiya gaya hai.

1. Tissue Basics — Cell > Tissue > Organ > System

- Tissue = similar origin/basic structure wale cells ka group jo ek unit ki tarah function karta hai
- Multicellular organisms mein tissues division of labour create karte hain
- Organisation level = Cell > Tissue > Organ > Organ System > Animal
- Plant tissues = Meristematic + Permanent
- Animal tissues = Epithelial + Connective + Muscular + Nervous

TRICK - “ECMN = Epithelial - Connective - Muscular - Nervous”

Animal tissue ke 4 types ka direct code.

2. Meristematic Tissue — A-I-L

- Immature/undifferentiated cells; continuously divide
- No intercellular spaces, no vacuoles
- Prominent nucleus + dense cytoplasm + thin cellulose wall

Meristem	Location	Function
Apical	Shoot/root tips	Length / elongation
Intercalary	Leaf base, nodes/internodes	Length growth
Lateral	Sides of stem/root	Thickness / girth

TRICK - “A-I-L = Apex Length, Inter-node Length, Lateral Girth”

Apical = tip; Intercalary = node/leaf base; Lateral = side.

EXAM TRAP -

Protoderm = apical meristem ka root-associated part, as stated in source notes.

TRICK - “Cork = Secondary Meristem Product”

Source notes cork ko secondary meristematic tissue batate hain; cork cells dead + suberin.

3. Simple Permanent Tissue — P-C-S

Tissue	Living/Dead	Wall / Space	Main Role
Parenchyma	Living	Thin wall; spaces present	Storage
Collenchyma	Living	Corner thickening; spaces usually absent	Support + flexibility
Sclerenchyma	Dead	Thick lignified wall; spaces absent	Strength / safety

TRICK - “P-C Live, S Dead”

Parenchyma + Collenchyma living; Sclerenchyma dead.

EXAM TRAP - “Coconut Husk = Sclerenchyma”

Direct source example.

4. Parenchyma Modifications — Chloro vs Aero

Modification	Feature	Function
Chlorenchyma	Chloroplast present	Photosynthesis
Aerenchyma	Large air spaces	Buoyancy in aquatic plants

TRICK - “Chloro = Chlorophyll; Aero = Air”

Chlorenchyma photosynthesis; Aerenchyma floating.

EXAM TRAP -

Aerenchyma = buoyancy, not photosynthesis.

5. Complex Permanent Tissue — Xylem vs Phloem

Tissue	Transport	Key Components / Fact
Xylem	Unidirectional water + minerals	Tracheids, vessels, xylem parenchyma, fibres
Phloem	Bidirectional food	Food-conducting complex tissue

TRICK - “Xylem = X-it Upward; Phloem = Food Both Ways”

Source note: Xylem unidirectional roots > stem > leaves.

Phloem bidirectional food transport.

EXAM TRAP -

Xylem parenchyma = living component of xylem.

6. Protective Tissue — Epidermis + Cork

Feature	Epidermis	Cork
Nature	Outermost single layer	Multilayered dead cells
Special	Cuticle may form waxy layer	Suberin in walls
Intercellular space	Not highlighted	Absent
Extra	Root hairs from root epidermis	Waterproof protection

TRICK - “Cork = Dead + Suberin + No Space = Waterproof”

Mechanical injury, infection aur water loss se protection.

TRICK - “Root Hair = Epidermis Extension”

Water absorption ke liye surface area badhata hai.

7. Stomata — Dicot Bean, Monocot Dumbbell

- Stomata = leaf epidermis ke small pores
- Functions = transpiration + gaseous exchange
- Water enters guard cells > turgid > stoma opens
- Water exits > relaxed/shrunken > stoma closes
- Dicot guard cells = bean-shaped
- Monocot guard cells = dumbbell-shaped

TRICK - “D-B / M-D = Dicot-Bean, Monocot-Dumbbell”

Guard-cell shape ka direct exam pair.

EXAM TRAP -

Water IN = Open; Water OUT = Close.

8. Epithelial Tissue — SC3 + Stratified

Type	Location
Simple squamous	Blood vessels, alveoli, mouth, alimentary canal
Stratified squamous	Skin
Cuboidal	Kidney tubules, sweat/salivary/thyroid glands
Columnar	Stomach, intestine, gall bladder, bile duct
Ciliated columnar	Trachea, oviduct, vas deferens
Glandular single-cell	Goblet cell

TRICK - “SC3 = Squamous - Cuboidal - Columnar - Ciliated Columnar”

Source quick-recall code for simple epithelial forms.

EXAM TRAP - “Kidney = Cuboidal; Trachea = Ciliated; Skin = Stratified”

3 highest-yield location traps.

9. Tendon vs Ligament — M vs B

Tissue	Connects	Fibre / Flexibility
Tendon	Bone to Muscle	White collagen; strong, less flexible
Ligament	Bone to Bone	Yellow elastin; more flexible

TRICK - “Tendon = To Muscle; Ligament = Links Bones”

Tendon B-M; Ligament B-B.

EXAM TRAP -

Collagen = Tendon; Elastin = Ligament.

10. Blood & Lymph — 55/45 + RBC Journey

Component	Key Fact
Plasma	55%; about 90% water, 8% protein, 2% minerals/ions
RBC	Formed bone marrow; destroyed spleen
WBC	Neutrophil, Eosinophil, Basophil, Lymphocyte, Monocyte
Platelets	Blood clotting
Lymph	Colourless; RBC absent

TRICK - “55 Plasma + 45 Cells”

Source blood ratio.

TRICK - “RBC: Bone Marrow Birth, Spleen Burial”

Spleen = graveyard of RBC.

TRICK - “N-E-B-L-M”

WBC types: Neutrophil-Eosinophil-Basophil-Lymphocyte-Monocyte.

EXAM TRAP -

Lymph = colourless + RBC absent.

11. Loose Connective — Adipose vs Areolar

Tissue	Where / Nature	Function
Adipose	Below skin, around organs, bone marrow	Fat storage + insulation + shock absorption
Areolar	Loose jelly matrix; below skin / spaces / near muscles-vessels	Repair + filling spaces

TRICK - “Adipose = Add Fat; Areolar = Area Fill”

Adipose stores; Areolar repairs/fills.

12. Bone vs Cartilage — Hard Salts vs Chondrin

Tissue	Matrix / Nature	High-Yield Facts
Bone	Hard/stiff; CaPO_4 + CaCO_3	Osteoblasts = bone-forming; support/protection
Cartilage	Flexible; chondrin + sugar + collagen + elastin	Joints, nose tip, outer ear, trachea, throat

TRICK - “Bone = CaPO_4 + CaCO_3 ; Cartilage = Chondrin”

Hard vs flexible ko matrix ke saath pair karo.

TRICK - “Osteoblast = Builds Bone”

Stem cells source notes mein bone marrow mein present.

EXAM TRAP -

Bone salt pair ko CaPO_4 + CaCO_3 se yaad rakho.

13. Muscle — Striated / Smooth / Cardiac

Feature	Striated	Smooth	Cardiac
Shape	Cylindrical	Spindle	Branched cylindrical
Nuclei	Multi	Uni	Uni
Bands	Present	Absent	Present
Control	Voluntary	Involuntary	Involuntary
Fatigue	Fast	Slow/resistant	No fatigue
Special	Skeleton attached	Internal organs	Intercalated discs

TRICK - “Bands = Striated + Cardiac; Voluntary = Striated Only”

Smooth no bands; Cardiac striated but involuntary.

TRICK - “Cardiac = Branched + Intercalated + No Fatigue”

Heart muscle ka signature trio.

EXAM TRAP -

Cardiac striated hote hue bhi involuntary hai.

14. Muscle Proteins — A-M

- Muscle fibres = elongated contractile cells
- Movement by contraction-relaxation
- Contractile proteins = Actin + Myosin

TRICK - “A-M = Actin + Myosin = Action of Muscle”

Muscle movement ka direct protein pair.

15. Nervous Tissue — Neuron + Neurilemma

- Functions = sensitivity, excitability, coordination
- Neuron = structural + functional unit
- Neuron length source note = up to about 1 m
- Neurons lack dividing capability
- Locations = brain, spinal cord, nerves
- Schwann cells = Neurilemma cells in source notes

TRICK - “Neuron = Nerve Unit, No Division”

Neuron structure/function unit + non-dividing.

EXAM TRAP - “Schwann = Neurilemma”

Source-specific terminology direct pair.

16. Additional One-Liners

- Eosine = luminous red acidic dye
- Eosine stains cytoplasm, RBC, collagen and muscle fibres
- Stem cells present in bone marrow
- Cork cells are dead and contain suberin
- Robert Hooke observed cork cells and coined “Cellula”

TRICK - “Eosine = Red Acidic Dye”

Cytoplasm + RBC + collagen + muscle fibres ko stain karta hai.

17. Final Exam Traps — 12 Red Flags

- Apical tips/length; Intercalary nodes-leaf base/length; Lateral sides/thickness
- Parenchyma + Collenchyma living; Sclerenchyma dead
- Aerenchyma buoyancy; Chlorenchyma photosynthesis
- Xylem unidirectional water; Phloem bidirectional food
- Dicot guard = bean; Monocot = dumbbell
- Cork = dead + suberin + no intercellular space
- Cuboidal kidney; Ciliated trachea/oviduct/vas deferens; Skin stratified squamous
- Tendon bone-muscle/collagen; Ligament bone-bone/elastin
- RBC formed marrow, destroyed spleen
- Bone = CaPO_4 + CaCO_3 ; Cartilage = chondrin
- Cardiac = striated + involuntary + branched + intercalated discs + no fatigue
- Neuron does not divide; Schwann = Neurilemma

EXAM TRAP - “A-I-L / P-C-S / Chloro-Aero / X-P / D-B-M-D / Cork / Epi / T-L / RBC / Bone / Cardiac / Neuron”

Chapter ke 12 most-confusable zones isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Tissue = similar cells; Plant M+P, Animal ECMN → Meristem A-I-L: Apical tip-length, Intercalary node-length, Lateral side-girth; no vacuole/space → Simple P-C-S = Parenchyma live/storage, Collenchyma live/flexible, Sclerenchyma dead/strength; Coconut=Sclerenchyma; Chloro photosynthesis, Aero buoyancy → Xylem one-way water, Phloem two-way food, xylem parenchyma living → Epidermis + Cork; Cork dead+suberin+no space; Root hair epidermis → Stomata: Dicot bean, Monocot dumbbell; water-in open, water-out close → Epithelial SC3; kidney cuboidal, trachea ciliated, skin stratified → Tendon B-M collagen vs Ligament B-B elastin → Blood 55 plasma/45 cells; RBC marrow birth-spleen burial; WBC N-E-B-L-M; lymph colourless/no RBC → Adipose fat, Areolar fill; Bone $\text{CaPO}_4 + \text{CaCO}_3$ /Osteoblast vs Cartilage Chondrin → Muscle Actin+Myosin; bands Striated+Cardiac, voluntary Striated only; Cardiac branched/intercalated/no fatigue → Neuron functional unit/no division; Schwann=Neurilemma; Eosine red acidic dye.”

TOP 5: A-I-L meristems | P-C living vs S dead | Xylem one-way vs Phloem two-way | Tendon B-M vs Ligament B-B | Cardiac striated but involuntary.

BIOLOGICAL PROCESS — MEMORY TRICK SHEET

General Science · Biology · Chapter 6 · Nutrition + Digestion + Respiration + Blood + Excretion + Plant Transport

Ye full notes nahi hai — source chapter ke high-memory facts, numbers, enzyme-location pairs, vitamin deficiencies, heart/blood exceptions, nephron path aur plant transport ko exam-recall tricks mein compress kiya gaya hai.

1. Life Processes — N-R-E-T

Process	Meaning
Nutrition	Getting nutrients from food
Respiration	Energy release from food
Excretion	Removal of metabolic wastes
Transportation	Moving substances in body

TRICK - “N-R-E-T = Nutrition - Respiration - Excretion - Transport”

Biological process chapter ka master four-process code.

2. Plant Nutrition — Auto + Special Modes

- Autotrophic nutrition = inorganic substances se organic food banana
- Photosynthesis chlorophyll ki presence mein hota hai
- Source equation: $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{sunlight} > \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$
- Plant macronutrients listed = C, H, O, N, P, K, Ca, Mg, S
- Micronutrients listed = Fe, B, Zn, Mn, Cu, Mo, Cl

TRICK - “Auto = Apna Food; Hetero = Har Baar Bahar”

Autotroph khud food banata; heterotroph organic source par depend karta.

Special Mode	Examples / Hook
Insectivorous	Nepenthes, Drosera, Venus flytrap - N deficient soil
Parasitic	Cuscuta - complete stem parasite; sucker roots
Symbiotic	Legume root nodules + Rhizobium - N fixing

TRICK - “N-D-V / Cuscuta Sucker / Rhizobium Root-N”

Insectivorous = Nepenthes-Drosera-Venus.

Cuscuta parasite; Rhizobium symbiotic N-fixer.

EXAM TRAP -

Cuscuta = parasite; Rhizobium = symbiont, parasite nahi.

3. Heterotrophic Modes — H-S-P

Mode	Process	Examples
Holozoic	Ingest > Digest > Absorb	Human, Amoeba, Goat, Lion
Saprophytic	Dead/decayed matter ka extracellular digestion	Rhizopus, Yeast, Penicillium
Parasitic	Host ke inside/outside	Ticks, Lice, Leech, Tapeworm, pathogenic bacteria

TRICK - “H-S-P = Holozoic - Saprophyte - Parasite”

Holo = ingest; Sapro = dead matter; Para = host.

4. Energy Values — 9.3 > 4.3 > 4.1

Nutrient	Energy / g	High-Yield Fact
Fat	9.3 Kcal	Highest energy

Protein	4.3 Kcal	Deficiency: Marasmus + Kwashiorkor
Carbohydrate	4.1 Kcal	C:H:O = 1:2:1

TRICK - “F-P-C = 9.3 - 4.3 - 4.1”

Fat > Protein > Carbohydrate energy order.

TRICK - “Protein kam = M-K”

M = Marasmus; K = Kwashiorkor.

- Term “Vitamin” coined by Casimir Funk
- Body water in source notes = about 65-70%
- Essential amino acids = His, Ile, Leu, Lys, Met, Phe, Thr, Trp, Val

TRICK - “HILL-MPTTV”

Essential amino acids mnemonic chunk: His-Ile-Leu-Lys / Met-Phe-Thr-Trp-Val.

5. Vitamins — ADEK vs B+C

TRICK - “ADEK = Fat Soluble; B+C = Water Soluble”

Solubility ka biggest vitamin trap.

Vitamin	Chemical Name	Deficiency / Effect
A	Retinol / beta-Carotene	Night blindness, xerophthalmia
D	Calciferol	Rickets child; osteoporosis adult
E	Tocopherol	Sterility/infertility; weak muscles
K	Naphthoquinone/Phylloquinone	Bleeding due to clot failure

TRICK - “A-Night, D-Rickets, E-Sterility, K-Klotting”

ADEK ko deficiency/function se pair karo.

Vitamin	Name	Deficiency
B1	Thiamine	Beri-Beri
B2	Riboflavin	Cheilosis
B3	Niacin	Pellagra - 3D
B5	Pantothenic acid	Skin/growth/hair/fertility effects
B6	Pyridoxine	Anaemia, skin disease, cramps
B7	Biotin	Skin disease, hair loss
B9	Folic acid	Anaemia, growth, neural-tube defects
B12	Cobalamin	Anaemia + nervous/RBC/DNA effects
C	Ascorbic acid	Scurvy

EXAM TRAP - “B1 Beri, B3 3D, C Scurvy, D Rickets, K Klotting”

B3 Pellagra = Dermatitis + Diarrhoea + Dementia.

6. Digestion Process — I-D-A-A-D

- Ingestion > Digestion > Absorption > Assimilation > Defecation
- Alimentary canal length in source notes = 8-10 m

TRICK - “I-D-A-A-D”

Food journey ka 5-step process.

Part	Source Length / pH	Core Function
Pharynx	12 cm	Food-air common passage; epiglottis protects trachea
Oesophagus	22-24 cm	Peristalsis
Stomach	25 cm; pH 2.0	Protein digestion + chyme

Small intestine	25 cm + 2.4 m + 3.5 m	Complete digestion + microvilli absorption
Large intestine	Caecum 6 cm + Colon 1.3 m + Rectum 12-15 cm	Water absorption + faeces

TRICK - “Stomach 2; Small Complete; Large Water”

pH 2 stomach, complete digestion small intestine, water absorption large intestine.

7. Teeth — 2-1-2-3 = 32

- Adult dental formula per half jaw = 2 incisors, 1 canine, 2 premolars, 3 molars
- Total adult teeth = 32
- Child formula = 2,1,0,2
- Wisdom tooth = third molar
- Enamel = outer layer and hardest; Dentine = inner layer

TRICK - “2-1-2-3 x 4 = 32”

Adult formula ka direct arithmetic trick.

Tongue Zone - Source Note	Taste
Front	Sweet
Front-side	Salty
Side	Sour
Back	Bitter

8. Digestive Enzymes — Mouth-Stomach-Pancreas-Intestine

Location	Key Enzymes	Acts On
Mouth	Ptyalin / Amylase	Starch
Stomach	Pepsin, Rennin, Gastric lipase	Protein, milk protein, fat
Pancreas	Trypsin, Chymotrypsin, Amylopsin, Carboxypeptidase, Lipase	Protein/starch/polypeptide/fat
Intestine	Enterokinase, Erepsin, Maltase, Sucrase, Lactase, Lipase, Nuclease	Final digestion targets

TRICK - “Mouth A; Stomach P-R-L; Pancreas T-C-A-C-L”

Mouth = Amylase.

Stomach = Pepsin-Rennin-Lipase.

Pancreas = Trypsin-Chymotrypsin-Amylopsin-Carboxypeptidase-Lipase.

EXAM TRAP -

Salivary Ptyalin/Amylase acts on starch.

9. Liver + Pancreas — Largest vs Second Largest

- Liver = largest gland
- Kupffer cells = macrophages
- Bile emulsifies fat; source pH 7.4-8.5
- Bile stored in gall bladder
- Bile pigments = bilirubin + biliverdin
- Liver functions include glycogenesis, detoxification, urea formation, storage of A,D,E,K
- Pancreas = second largest gland; mixed gland; pancreatic juice
- Salivary glands = 3 pairs
- Source note states: Gall stones = Calcium oxalate

TRICK - “Liver Largest; Pancreas 2nd + Mixed”

Largest-vs-second-largest direct trap.

TRICK - “B-B = Bilirubin + Biliverdin; Bile in Gall Bladder”

Pigments + storage ek saath yaad karo.

10. Respiration — 36 vs 2 ATP

Feature	Aerobic	Anaerobic
O ₂	Required	Not required
Breakdown	Complete	Partial
Energy	36 ATP	2 ATP
Location	Cytoplasm + mitochondria	Cytoplasm only
Products	CO ₂ + H ₂ O	Ethanol + CO ₂ in yeast
Examples	Most animals/plants	Yeast, bacteria, exercising human muscle

TRICK - “AERO = Air + 36; ANAERO = Absent Air + 2”

Energy difference is the key exam number.

EXAM TRAP -

Human muscles during exercise = anaerobic; yeast fermentation = anaerobic.

11. Human Respiratory System — 3 Right, 2 Left

Part	High-Yield Fact
Nostrils	Mucus + hairs clean/moisten air
Pharynx	Common passage; epiglottis
Larynx	Sound; Adam’s apple in men
Trachea	10-12 cm; C-shaped hyaline cartilage rings
Right lung	Large/broad; 3 lobes + 3 bronchi
Left lung	Small/thin; 2 lobes + 2 bronchi
Alveoli	Balloon-shaped, single-layered; gas exchange

TRICK - “Right 3, Left 2; Alveoli = Exchange”

Lung lobes ka direct pair.

Phase	O ₂	CO ₂	Diaphragm
Inspiration	21%	0.03%	Flattens downward
Exhalation	16%	4.4%	Dome-shaped

TRICK - “Inhale Flat; Exhale Dome”

Diaphragm shape trick.

12. Gas Transport — O₂ 97; CO₂ 70-23-7

- O₂ transport = 97% by haemoglobin
- CO₂ = 70% bicarbonate + 23% haemoglobin + 7% plasma

TRICK - “O₂ = 97 Hb; CO₂ = 70-23-7”

Gas-transport number pack.

13. Respiratory Surfaces — Membrane-Skin-Gill-Trachea-Lung

Organism	Surface
Amoeba & Planaria	Cell membrane
Earthworm & Leech	Moist skin
Fish, Prawns, Tadpole	Gills
Frog	Moist skin + lungs

Mosquito & Housefly	Trachea + spiracles
Birds/Lizards/terrestrial animals	Lungs

TRICK - “Amoeba Membrane, Worm Skin, Fish Gill, Insect Trachea, Land Lung”

Frog exception = moist skin + lungs.

EXAM TRAP -

Insects = trachea + spiracles, not gills.

14. Blood — 55/45, pH 7.3-7.5

- Blood pH = 7.3-7.5, alkaline
- Adult blood = 5-6 L, about 7-8% body weight
- Plasma = 55%; cells = 45%

TRICK - “Blood = 55 Liquid + 45 Cells”

Plasma-cell composition split.

Cell	Count / Lifespan	Formation / Role
RBC	4.5-5 million/mm ³ ; 120 days	Bone marrow after birth; gas transport
WBC	6,000-8,000/mm ³ ; hours-months	Immunity
Platelets	1.5-3.5 lakh/mm ³ ; 8-10 days	Megakaryocytes; clotting

TRICK - “RBC 120; Platelet 8-10; Neutrophil 60-65”

Blood-cell exam numbers.

TRICK - “RBC Birth = Marrow; Graveyard = Spleen”

Formation-destruction pair from source.

15. WBC Types — N-E-B / L-M

- Granular = Neutrophils 60-65%, Eosinophils 2-3%, Basophils 0.5-1%
- Agranular = Lymphocytes 20-25%, Monocytes 2-10%
- Neutrophils = most abundant WBC

TRICK - “Granular N-E-B; Agranular L-M”

N-E-B / L-M classification.

16. Blood Groups — L / Castello-Sturli / Rh

- A, B, O discovered by Landsteiner
- AB discovered by Von de Castello & Sturli, 1902
- Source notes: O = Universal Donor; AB = Universal Recipient
- Rh factor discovered by Landsteiner & Wiener, 1940 in Rhesus monkey
- Source notes: Rh+ = 85% world, 97% India

TRICK - “Landsteiner ABO; C-S AB; L-W Rh 1940”

Discoverer chain ko 3 blocks mein yaad karo.

EXAM TRAP -

Source note uses O = universal donor and AB = universal recipient.

17. Heart — 4 Chambers + Valve Side Trick

- Heart = 4 chambered: 2 atria + 2 ventricles

- Atria receive blood; ventricles pump
- Double-walled pericardium covers heart
- Source weights: men 280-340 g; women 230-240 g; newborn 20 g

Feature	Fact
Tricuspid	Right atrium - right ventricle
Bicuspid / Mitral	Left atrium - left ventricle
Semilunar	Ventricle - artery
Lub	Closure of AV valves
Dup	Closure of semilunar valves
Heart rate	72-75 beats/min
One beat	0.85 s
Cardiac output	5 L/min
Blood pressure	120/80 mmHg
Pacemaker	SA Node

TRICK - “TRI = Right; BI/Mitral = Left”

Tricuspid right, bicuspid/mitral left.

TRICK - “Lub AV; Dup Semi”

Heart sound valve-closure trick.

TRICK - “72-75 / 0.85 / 5 L / 120-80 / SA”

Heart number pack.

18. Blood Flow — Right Deoxy, Left Oxy

Chamber	Flow
Right atrium	Deoxygenated blood from vena cava
Right ventricle	Deoxygenated to lungs via pulmonary artery
Left atrium	Oxygenated from lungs via pulmonary veins
Left ventricle	Oxygenated to body via aorta

TRICK - “RA Receive Deoxy; RV to Lungs; LA Receive Oxy; LV to Body”

Heart circulation ko R-R-L-L chain mein bolo.

EXAM TRAP - “Pulmonary Exception = Artery Deoxy, Vein Oxy”

Major vessel exception.

19. Artery vs Vein vs Capillary

Vessel	Direction / Blood	Wall / Valve
Artery	Away; usually oxygenated	Thick; no valves
Vein	Toward; usually deoxygenated	Thin; valves present
Capillary	Connect arteries-veins	Exchange surface

TRICK - “A = Away; V = Visits Heart”

Pulmonary vessels ko exception ke roop mein separate yaad karo.

- Varicose veins = dilated superficial veins due to incompetent venous valves, commonly legs

20. Excretion — Plant + Human

Plant Waste	Method
Liquid	Transpiration via stomata
Gases	Stomata + lenticels
Solid	Vacuoles/dead tissues; leaf shedding

Useful waste	Oils, perfumes, gums, resins, rubber, tannin
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TRICK - “Plant Waste = Stoma-Lenticel-Vacuole-Shedding”

Plant excretion ke routes.

- Human nitrogenous waste mainly urea; urea produced in liver and removed by kidneys
- Kidneys = one pair, bean-shaped
- About 10 lakh nephrons per kidney
- Nephron = structural + functional unit
- Urine pH about 6.0, slightly acidic
- Urine source composition = water 95% + urea 2%
- Urine colour = urochrome/urobilin

TRICK - “Liver Makes Urea; Kidney Removes Urea”

Formation vs excretion organ trap.

21. Urinary Path + Nephron Path

Organ	Role
Kidney	Forms urine
Ureter	Kidney > bladder
Bladder	Stores urine
Urethra	Removes urine

TRICK - “K-U-B-U = Kidney-Ureter-Bladder-Urethra”

Urinary tract order.

TRICK - “B-P-H-D-C”

Bowman’s capsule > PCT > Henle loop > DCT > Collecting duct.

TRICK - “B-P-H-D-C > Ureter > Bladder > Urethra”

Source chapter ka complete nephron-to-exit chain.

EXAM TRAP -

Nephron = structural & functional unit; about 10 lakh per kidney.

22. Plant Transport — Xylem vs Phloem

Transport	Tissue / Material	Direction	Mechanism
Water	Xylem; water + minerals	Upward / unidirectional	Cohesion + adhesion + transpiration pull
Food	Phloem; sucrose	Bi-/multidirectional	Mass flow - Munch hypothesis

TRICK - “Xylem = Water Up; Phloem = Food Both”

X = xylem-water; P = phloem-food.

TRICK - “Munch = Mass Flow”

Food translocation hypothesis.

EXAM TRAP -

Xylem unidirectional; phloem bidirectional/multidirectional.

23. Final Exam Traps — 14 Red Flags

- Cuscuta parasite vs Rhizobium symbiotic N-fixer
- Fat 9.3 > Protein 4.3 > Carbohydrate 4.1 Kcal/g
- Protein deficiency = Marasmus + Kwashiorkor
- ADEK fat-soluble; B-complex + C water-soluble
- B1 Beri-Beri; B3 Pellagra; C Scurvy; D Rickets; K bleeding
- Liver largest gland; pancreas second largest + mixed
- Aerobic 36 ATP vs anaerobic 2 ATP
- Frog skin+lungs; insects trachea+spiracles
- RBC 120 days; platelets 8-10 days; neutrophils most abundant
- Source notes: O universal donor; AB universal recipient
- Tricuspid right; bicuspid/mitral left; semilunar ventricle-artery
- Pulmonary artery deoxygenated; pulmonary vein oxygenated
- Nephron structural-functional unit; urine pH about 6
- Xylem water one-way; phloem food both ways

EXAM TRAP - "Nutrition-Energy-Vitamin-Glands-ATP-Gas-Blood-Valves-Pulmonary-Kidney-X/P"

Exam ke major confusion zones isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"NRET life processes → Auto chlorophyll/photosynthesis; N-D-V insectivorous, Cuscuta sucker parasite, Rhizobium root-N symbiotic; hetero H-S-P → energy F-P-C 9.3-4.3-4.1; protein M-K; Vitamin Casimir Funk; ADEK fat vs B+C water; B1 Beri, B3 3D Pellagra, C Scurvy, D Rickets, K Klotting → digestion I-D-A-A-D; canal 8-10m; teeth 2-1-2-3x4=32; Mouth Amylase, Stomach P-R-L, Pancreas T-C-A-C-L; Liver largest/bile in gall bladder, Pancreas 2nd mixed → respiration Aero36 vs Anaero2; Right lung3/Left2; inhale-flat/exhale-dome; O₂=97Hb, CO₂=70-23-7; Frog skin+lung, insect trachea → blood pH7.3-7.5, 55 plasma/45 cells; RBC120 marrow-birth/spleen-grave; platelets8-10 megakaryocyte; WBC N-E-B/L-M → Landsteiner ABO, Castello-Sturli AB, L-W Rh1940 → heart4; TRI-right, BI-left, LubAV/DupSemi, SA node, 120/80 → RA deoxy in, RV pulmonary artery out; LA oxy in, LV aorta out; pulmonary artery deoxy/vein oxy → excretion Liver makes urea/Kidney removes; K-U-B-U; nephron B-P-H-D-C; urine pH6 → plants: Xylem water-up by transpiration pull; Phloem sucrose both-way by Munch mass flow."

TOP 5: Fat 9.3 > Protein 4.3 > Carb 4.1 | ADEK fat-soluble | Aero36 vs Anaero2 | Pulmonary artery deoxy / vein oxy | Xylem one-way water vs Phloem both-way food.

CONTROL, COORDINATION & MOVEMENT — MEMORY TRICK SHEET

General Science · Biology · Chapter 7 · Nervous System + Hormones + Muscles + Skeleton + Joints + Plant Coordination

Ye full notes nahi hai — source chapter ke impulse pathway, brain-function pairs, endocrine hormones, muscle/skeleton superlatives, joint examples aur plant movement/hormone traps ko fast exam recall ke liye compress kiya gaya hai.

1. Nervous System — C-P-A

Division	Components / Role
CNS	Brain + Spinal cord
PNS	12 pairs cranial nerves + 31 pairs spinal nerves
ANS	Sympathetic + Parasympathetic

TRICK - “C-P-A = Central - Peripheral - Autonomic”

PNS number pack = 12 cranial + 31 spinal pairs.

ANS pair = Sympathetic + Parasympathetic.

2. Impulse Path — S-R-S-C-M-E

- Stimulus > Receptor > Sensory neuron > Brain/Spinal cord > Motor neuron > Effector organ
- Reflex action = automatic + involuntary response
- Source notes associate reflex control with spinal cord

TRICK - “S-R-S-C-M-E”

Stimulus-Receptor-Sensory-CNS-Motor-Effector.

EXAM TRAP -

Reflex = automatic/involuntary; source notes mein spinal cord se associated, brain se nahi.

3. Sensory Receptors + Eye

Organ	Function
Ear	Hearing + body balance
Eye	Vision
Skin	Touch + heat + pain
Nose	Smell
Tongue	Taste buds

Eye Part	Fact
Iris	Controls pupil expansion/contraction
Cornea	Transparent front part
Lacrimal gland	Releases tears
Superior rectus	Controls eye movement

TRICK - “I-C-L-S = Iris-Cornea-Lacrimal-Superior Rectus”

Iris pupil, Cornea transparent, Lacrimal tears, Superior rectus movement.

EXAM TRAP -

Nervous coordination = electrical impulses; endocrine coordination = chemical hormones.

4. Brain — Fore-Mid-Hind

- Brain weight in source notes = about 1300-1400 g

- Forebrain = higher functions + voluntary actions
- Midbrain = visual + auditory reflexes
- Hindbrain = automatic functions + spinal coordination

TRICK - “FORE Thinks, MID Reflexes, HIND Runs Basics”

Fore = higher functions; Mid = light/sound reflex; Hind = automatic coordination.

5. Forebrain — O-C-E-H

Part	Function
Olfactory lobe	Smell
Cerebrum	Memory, consciousness, reasoning, learning
Epithalamus	Pain, sensation, heat/cold perception
Hypothalamus	Appetite, thirst, affection, disgust, sleep, temperature

TRICK - “O-C-E-H”

Olfactory-Smell, Cerebrum-Memory, Epithalamus-Sensation, Hypothalamus-Home regulation.

EXAM TRAP - “Cerebrum = Memory; Cerebellum = Balance”

Naam similar, function opposite trap.

6. Hindbrain — P-C-M

Part	Function
Pons	Respiratory centre
Cerebellum	Muscular balance + posture
Medulla oblongata	Breathing rhythm + heart rate + hiccups + ANS

TRICK - “P-C-M = Pons Respire, Cerebellum Balance, Medulla Heart/Breath”

Hindbrain ke 3 direct function pairs.

EXAM TRAP -

Medulla = breathing + heart rate; Hypothalamus = appetite/thirst/sleep/temp.

7. Endocrine Glands — Hormone-Disease Pairs

Gland / Cell	Hormone	Main Function / Deficiency
Anterior Pituitary	GH / Somatotrophic	Growth; deficiency = dwarfism/stunted growth
Thyroid	Thyroxine	BMR; deficiency linked with cretinism/goitre
Adrenal	Adrenaline / Nor-adrenaline	Raises BP/heartbeat; source calls adrenaline Do or Die
Pancreas beta-cells	Insulin	Lowers blood sugar; deficiency = diabetes
Pancreas alpha-cells	Glucagon	Raises blood sugar
Testes	Testosterone / Androgen	Male secondary characters + spermatogenesis
Ovaries	Oestrogen + Progesterone	Female secondary characters, oogenesis, uterus prep

TRICK - “Beta Insulin DOWN; Alpha Glucagon UP”

beta-cells lower blood sugar; alpha-cells raise blood sugar.

TRICK - “GH-Dwarf | Thyroid-Goitre/Cretinism | Insulin-Diabetes”

Disease-hormone fast pair.

EXAM TRAP - “Adrenaline = Do or Die”

Source ka direct one-liner.

8. Muscles — V-I-I + Superlatives

Feature	Striated/Skeletal	Smooth	Cardiac
Control	Voluntary	Involuntary	Involuntary
Bands	Present	Absent	Present
Location	Skeleton	Uterus/intestine/pupil/trachea	Heart

TRICK - “V-I-I; Bands = Skeletal + Cardiac”

Striated voluntary; Smooth involuntary; Cardiac involuntary but striated.

TRICK - “Masseter Strongest; Gluteus Maximus Largest”

Muscle superlatives.

EXAM TRAP -

Cardiac muscle striated hote hue bhi involuntary hai.

9. Skeleton — 206 + Head 8-14-6-1

- Total adult bones = 206
- Cranium = 8
- Face = 14
- Ear ossicles = 6
- Hyoid = 1

TRICK - “Head = 8 + 14 + 6 + 1”

Cranium-Face-Ear-Hyoid.

TRICK - “Ear = M-I-S; Stapes Smallest”

Malleus-Incus-Stapes; Stapes = smallest bone.

10. Vertebral Column — 7-12-5-1-1

Region	Adult Count
Cervical	7
Thoracic	12
Lumbar	5
Sacrum	1
Coccyx	1

TRICK - “7-12-5-1-1 = 26 Adult Vertebral Bones”

Neck-Chest-Waist-Sacrum-Coccyx.

EXAM TRAP -

Source notes mention sacrum 5 and coccyx 4 in infants, but adult table counts them as 1 each.

11. Skeleton Superlatives + Limb Numbers

- Largest & strongest bone = Femur
- Smallest bone = Stapes
- Ribs = 24 = 12 pairs

- Sternum = 1
- Pectoral girdle = Scapula 2 + Clavicle 2 = 4
- Pelvic girdle = Innominate 2
- Upper arm = Humerus 2; Lower arm = Radius 2 + Ulna 2
- Wrist/Carpals = 16
- Palm/Fingers = Metacarpals 10 + Phalanges 28
- Lower limb: Femur 2; Tibia 2 + Fibula 2; Patella 2; Tarsal 14 + Metatarsals 10; toe phalanges 28

TRICK - “Femur Biggest-Strongest; Stapes Smallest; Ribs 24”

3 skeletal super-facts.

12. Joints — S-A-E-S

Joint Type	Mobility	Source Example
Immovable	No movement	Skull bones; teeth & jaw joint
Semi-movable	Limited	Pubic symphysis / pelvic girdle
Ball & Socket	All directions	Shoulder; source table also lists ankles/feet
Hinge	One-direction	Elbow, fingertips
Pivotal	Rotation	Atlas-Axis neck
Saddle	Two-direction	Source table: wrist & ankle

TRICK - “Skull None - Atlas Axis Pivot - Elbow Hinge - Shoulder Ball”

Most-tested examples ka four-point chain.

TRICK - “Synovial = Smooth Movement”

Movable joints mein synovial cavity/fluid lubrication.

EXAM TRAP -

Ball & socket maximum movement; hinge mainly flexion/extension.

13. Plant Movement — Tropic vs Nastic

Movement	Nature	Examples
Tropic	Growth-related; direction matters	Photo, Geo, Hydro, Chemo
Nastic	Non-growth; direction-independent	Touch-me-not / Mimosa; thigmonasty

TRICK - “Tropic = Turns Toward/Away; Nastic = No Direction”

Tropic mein direction matters; nastic mein direction independent.

TRICK - “P-G-H-C”

Phototropism-Light, Geotropism-Gravity, Hydrotropism-Water, Chemotropism-Chemical.

- Source note: phototropism root = negative / away from light; sunflower example
- Geotropism: roots positive, shoots negative
- Hydrotropism: movement towards water
- Chemotropism: pollen tube grows towards ovule

EXAM TRAP -

Touch-me-not = nastic/thigmonasty, tropic nahi.

14. Plant Hormones — AGC vs AE

Category	Hormone	Key Function
Growth promoter	Auxin	Apical dominance + cell elongation + parthenogenesis
Growth promoter	Gibberellin	Bolting + seed germination
Growth promoter	Cytokinin	Cell division + delays senescence + breaks dormancy
Growth inhibitor	ABA	Stress hormone; stomatal closure + dormancy + senescence
Growth inhibitor	Ethylene	Gaseous hormone; fruit ripening + senescence

TRICK - “AGC = Auxin-Gibberellin-Cytokinin = Growth Promoters”

Source ka direct growth-promoter group.

TRICK - “AE = ABA-Ethylene = Growth Inhibitors”

ABA stress/dormancy; Ethylene fruit ripening.

TRICK - “Auxin Apex; Gibberellin Bolting; Cytokinin Cell Division”

A-G-C function hook.

EXAM TRAP -

Ethylene = only gaseous plant hormone in source notes; ABA = stress hormone.

15. Final Exam Traps — 11 Red Flags

- Reflex = automatic/involuntary; source notes associate it with spinal cord
- Nervous = electrical; Endocrine = chemical/hormonal
- Cerebrum memory vs Cerebellum balance
- Medulla breathing/heart rate vs Hypothalamus appetite/thirst/sleep/temp
- beta-cells insulin lowers sugar vs alpha-cells glucagon raises sugar
- Adrenaline = Do or Die hormone
- Cardiac = striated but involuntary; Masseter strongest; Gluteus maximus largest
- Skeleton = 206; Femur largest & strongest; Stapes smallest
- Vertebrae = 7-12-5-1-1; Ribs = 24
- Skull immovable; Atlas-Axis pivotal; Elbow hinge; Shoulder ball & socket
- Tropic growth/direction vs Nastic non-growth/direction-independent; Ethylene gaseous, ABA stress

EXAM TRAP - “Reflex-Electric/Chemical-Brain-Sugar-Muscle-Bones-Joints-Plant”

Chapter ke main confusion zones isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Nervous C-P-A = CNS brain+cord, PNS 12 cranial+31 spinal, ANS Sympathetic+Parasympathetic; impulse S-R-S-C-M-E; reflex spinal → receptors Ear balance/hear, Eye vision, Skin touch/heat/pain, Nose smell, Tongue taste; Iris pupil, Cornea clear, Lacrimal tears → brain Fore higher, Mid visual-auditory reflex, Hind automatic; Fore O-C-E-H, Hind P-C-M; Cerebrum memory vs Cerebellum balance, Medulla breath-heart, Hypothalamus hunger-thirst-sleep-temp → endocrine: GH dwarf, thyroid goitre/cretinism, adrenaline Do-or-Die, beta insulin DOWN, alpha glucagon UP, testes testosterone, ovaries oestrogen+progesterone → muscles V-I-I; cardiac striated+involuntary; Masseter strongest, Gluteus largest → skeleton 206; head 8-14-6-1; vertebrae 7-12-5-1-1; Femur biggest/strongest, Stapes smallest, ribs 24 → joints Skull none, Atlas-Axis pivot, Elbow hinge, Shoulder ball; synovial lubricates → plant movement Tropic direction/growth vs Nastic no-direction/non-growth; P-G-H-C → plant hormones AGC promoters = Auxin apex, Gibberellin bolting, Cytokinin division; AE inhibitors = ABA stress/dormancy + Ethylene gas/fruit ripening.”

TOP 5: Cerebrum memory vs Cerebellum balance | beta insulin DOWN vs alpha glucagon UP | Cardiac striated but involuntary | 7-12-5-1-1 vertebrae | Tropic direction vs Nastic no-direction.

REPRODUCTION — MEMORY TRICK SHEET

General Science · Biology · Chapter 8 · Asexual + Vegetative + Flower + Pollination + Human Reproduction

Ye full notes nahi hai — source chapter ke asexual methods, vegetative examples, flower parts, double fertilisation, pollination terms aur male-female reproductive system ke high-confusion facts ko fast recall format mein compress kiya gaya hai.

1. Asexual vs Sexual — Clone vs Variation

Feature	Asexual	Sexual
Parents	Single parent (source also notes bisexual)	Two parents of different sex
Gametes	Absent	Present
Fusion	No gamete fusion	Fertilisation present
Offspring	Clone; no variation	Variation present

TRICK - “A = Alone + Again Same; S = Sex Cells + Slightly Different”

Asexual = no gamete fusion + clone.

Sexual = gametes + fertilisation + variation.

EXAM TRAP -

Clone/no variation = asexual; variation = sexual.

2. Asexual Routes — F-F-R-B-S-V

- Fission
- Fragmentation
- Regeneration
- Budding
- Spore formation
- Vegetative reproduction

TRICK - “F-F-R-B-S-V”

Fission-Fragmentation-Regeneration-Budding-Spore-Vegetative.

3. Fission — Binary vs Multiple

Type	Result	Examples
Binary fission	2 progenies	Amoeba, Bacteria, Leishmania, Paramecium
Multiple fission	Many progenies	Plasmodium - malarial parasite

TRICK - “Binary = 2; Multiple = Many”

Leishmania binary; Plasmodium multiple.

EXAM TRAP -

Leishmania = binary fission; Plasmodium = multiple fission.

4. Fragmentation - Regeneration - Budding - Spores

Method	Examples
Fragmentation	Spirogyra, Rhizopus, Ulothrix, Red algae
Regeneration	Hydra, Planaria
Budding	Hydra, Yeast
Spore formation	Chlamydomonas, Rhizopus, Mucor, Puccinia

TRICK - “Hydra = R+B”

Hydra dono karta hai: Regeneration + Budding.

TRICK - “Yeast Buds; Plasmodium Multiplies”

Yeast = budding; Plasmodium = multiple fission.

EXAM TRAP -

Hydra ko sirf budding ya sirf regeneration mat yaad karo — source mein dono listed hain.

5. Tissue Culture — M-I-D

- Tissue culture = Micropropagation = In-vitro propagation
- Laboratory method
- Suitable culture medium
- Used for decorative + disease-free plants in less time

TRICK - “M-I-D = Micropropagation - In-vitro - Disease-free”

Teen words ek saath lock karo.

6. Natural Vegetative Propagation — S-R-L

Part	Examples
Stem	Potato, Sugarcane
Root	Rosewood, Sweet potato
Leaf	Begonia, Bryophyllum

TRICK - “Potato Stem - Sweet Potato Root - Bryophyllum Leaf”

Chapter ka most common vegetative trio.

EXAM TRAP -

Potato = stem; Sweet potato = root. Dono ko interchange mat karna.

7. Artificial Vegetative Propagation — Cutting + Layering

Method	Examples
Shoot cutting	Rose, Sugarcane, Banana
Root cutting	Lemon, Orange
Layering	Apple, Pear, Lemon, Strawberry

TRICK - “Shoot R-S-B; Root L-O; Layer A-P-L-S”

R-S-B = Rose-Sugarcane-Banana.

L-O = Lemon-Orange.

A-P-L-S = Apple-Pear-Lemon-Strawberry.

8. Flower Parts — C-C-A-G

Part	Memory
Calyx	Green part
Corolla	Coloured part
Androecium	Stamens - male organ
Gynoecium	Pistil/Carpel - female organ

TRICK - “C-C-A-G = Green - Colour - Male - Female”

Calyx green, Corolla colour, Androecium male, Gynoecium female.

TRICK - “A = Andro = Male; G = Gyno = Female”

Male-female floral organ pair.

9. Monoecious/Bisexual vs Dioecious — Maize Trap

- Source bisexual/monoecious examples = Hibiscus, Pea, Rose, Sunflower, Tomato, Mango, Lily
- Maize = monoecious plant but flowers are unisexual; male + female flowers separate on same plant
- Dioecious source examples = Papaya, Cucumber, Watermelon, Muskmelon

TRICK - “Maize = Same Plant, Separate Flowers”

Monoecious + unisexual flowers.

TRICK - “P-C-W-M = Separate Plants”

Papaya-Cucumber-Watermelon-Muskmelon = dioecious in source list.

EXAM TRAP -

Maize = monoecious but unisexual flowers; Papaya = dioecious.

10. Gametogenesis — Micro vs Mega

- Microsporogenesis = haploid pollen grains in anther
- Megasporogenesis = haploid embryo sac in ovule

TRICK - “Micro = Male Pollen; Mega = Female Embryo Sac”

Micro/anther vs Mega/ovule.

11. Double Fertilisation — $2n + 3n$

Fusion	Product
Male gamete n + Female gamete n	Zygote $2n$ > Embryo
Male gamete n + Polar/Secondary nuclei $2n$	Endosperm $3n$

TRICK - “1st = $2n$ Embryo; 2nd = $3n$ Endosperm”

Double fertilisation source notes mein Angiosperms ki uniqueness hai.

TRICK - “Ovule Seed; Ovary Fruit”

After fertilisation conversion pair.

EXAM TRAP -

Embryo = $2n$; Endosperm = $3n$. Ovule = seed; Ovary = fruit.

12. Pollination Types — A-G-C

Type	Meaning
Autogamy	Same bisexual flower
Geitonogamy	Different flowers of same plant
Cross-pollination	Different plants

TRICK - “Auto = Same Flower; Geito = Same Plant; Cross = Different Plant”

A-G-C ladder.

EXAM TRAP -

Source notes add: Geitonogamy genetically self-pollination-like.

13. Pollination Agents — A-H-E

Term	Agent	Examples
Anemophily	Wind	Maize, Bamboo, Sugarcane
Hydrophily	Water	Vallisneria, Hydrilla
Entomophily	Insects	Salvia, Fig, Rose

TRICK - “A-H-E = Air - H₂O - Entomo/Insect”

Anemo wind, Hydro water, Entomo insects.

EXAM TRAP -

Anemophily wind; Hydrophily water; Entomophily insects.

14. Male Reproductive System — L-S-E-V

Part / Cell	Function
Testes	Primary male organs; sperm + testosterone
Leydig cells	Testosterone
Sertoli cells	Nutrition/support of sperms
Epididymis	Sperm maturation
Vas deferens	Sperm transport

TRICK - “L = Leydig = Libido/Testosterone; S = Sertoli = Support”

Leydig testosterone; Sertoli sperm nutrition/support.

TRICK - “Epididymis = Matures; Vas = Moves”

Maturation vs transport.

- Testes located in scrotum, outside abdominal cavity
- Male accessory glands listed = Prostate, Cowper's, Perineal
- Source path components include Seminal vesicle > Urethra > Penis

EXAM TRAP -

Testes = scrotum outside abdomen. Leydig/Sertoli interchange mat karna.

15. Female Reproductive System — O-O-U

Part	Function / Fact
Ovaries	Primary female organs; eggs + female hormones
Oviducts	Pair; ovaries to uterus; carry egg
Uterus	Embryo development site

- Ovaries located in abdominal cavity
- Female hormones listed = Oestrogen + Progesterone
- Progesterone associated with corpus luteum in source notes
- Oogenesis = formation of haploid eggs
- Female accessory glands listed = Bartholin's + Perineal

TRICK - “O-O-U = Ovary - Oviduct - Uterus”

Egg source > transport > embryo-development site.

16. Male vs Female — Fast Pairing

Feature	Male	Female
Primary organ	Testes - scrotum	Ovaries - abdomen
Gamete formation	Spermatogenesis	Oogenesis
Gamete	Haploid sperm	Haploid egg
Hormone side	Leydig > Testosterone	Corpus luteum > Oestrogen/Progesterone in source table
Transport/Maturation	Epididymis / Vas deferens	Oviduct

TRICK - “Sperma = Sperm; Oo = Ovum”

Spermatogenesis male; Oogenesis female; both haploid through meiosis.

Fertilisation > diploid zygote 2n.

17. Final Exam Traps — 12 Red Flags

- Leishmania binary fission; Plasmodium multiple fission
- Hydra = budding + regeneration; Yeast = budding
- Tissue culture = micropropagation = in-vitro propagation
- Potato stem; Sweet potato root; Bryophyllum leaf
- Maize monoecious with unisexual flowers on same plant; Papaya dioecious
- Double fertilisation: embryo 2n; endosperm 3n
- Ovule > seed; Ovary > fruit
- Autogamy same flower; Geitonogamy same plant/different flower; Cross = different plants
- Anemophily wind; Hydrophily water; Entomophily insects
- Leydig testosterone; Sertoli sperm nutrition/support
- Spermatogenesis sperm; Oogenesis egg; both haploid
- Testes scrotum outside abdomen; ovaries abdominal cavity

EXAM TRAP - “Fission-Hydra-Culture-Veg-Maize-2n/3n-Seed/Fruit-AGC-AHE-LS-Gametes-Location”

12 source traps ko isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Asexual = alone/no fusion/clone vs Sexual = gametes+fertilisation+variation → routes F-F-R-B-S-V → Binary2: Amoeba-Bacteria-Leishmania-Paramecium; Multiple many: Plasmodium → Fragmentation Spirogyra/Rhizopus/Ulothrix/Red; Regeneration Hydra-Planaria; Budding Hydra-Yeast; Spores Chlamydomonas-Rhizopus-Mucor-Puccinia → Tissue culture = Micropropagation = In-vitro = disease-free → natural veg: Potato stem, Sweet potato root, Bryophyllum leaf; artificial Shoot R-S-B, Root L-O, Layer A-P-L-S → Flower C-C-A-G = Calyx green, Corolla colour, Andro male, Gyno female; Maize same plant separate unisexual flowers vs Papaya dioecious → Micro pollen/anther, Mega embryo-sac/ovule → Double fertilisation: 2n embryo + 3n endosperm; Ovule seed, Ovary fruit → Pollination A-G-C = same flower/same plant/different plants; agents A-H-E = wind/water/insects → Male: testes scrotum, Leydig testosterone, Sertoli support, Epididymis maturation, Vas transport → Female: ovaries abdomen, Oestrogen+Progesterone, Oogenesis, Oviduct transport, Uterus embryo → Spermatogenesis sperm + Oogenesis egg = haploid; fertilisation = zygote 2n.”

TOP 5: Leishmania binary vs Plasmodium multiple | Potato stem/Sweet potato root/Bryophyllum leaf | Maize monoecious vs Papaya dioecious | Embryo 2n vs Endosperm 3n | Leydig testosterone vs Sertoli support.

HEREDITY & EVOLUTION — MEMORY TRICK SHEET

General Science · Biology · Chapter 9 · Mendel + Crosses + DNA/RNA + Genetic Diseases + Evolution

Ye full notes nahi hai — source chapter ke Mendelian laws, F2 ratios, inheritance scientists, DNA/RNA facts, genetic disorders, evidences aur evolution theories ko fast exam-recall tricks mein compress kiya gaya hai.

1. Mendel Basics — M-P-7

- Gregor Johann Mendel = Father of Genetics
- Experiment plant = Pisum sativum / Garden pea
- Studied 7 pairs of contrasting characters
- Heredity = traits ka parent se offspring transmission
- Genetics = genetic traits aur inheritance ka study

TRICK - “M-P-7 = Mendel - Pisum - 7 Pairs”

Mendel ke opening facts ka one-line code.

TRICK - “T-R-Y-P vs D-W-G-W”

Dominant examples: Tall-Round-Yellow-Purple.

Recessive examples: Dwarf-Wrinkled-Green-White.

2. Mendel's 3 Laws — D-S-I

Law	Core Memory
Dominance	Dominant allele recessive ko mask karta hai
Segregation	Alleles gametogenesis mein separate
Independent Assortment	One trait ki inheritance dusre ko affect nahi karti

TRICK - “D-S-I = Dominance - Segregation - Independent”

Law 1, 2, 3 ka direct order.

TRICK - “Segregation = Purity of Gametes”

Source ka exact alternate name.

TRICK - “Mono = Law 1+2; Di = Law 3”

Monohybrid supports Dominance + Segregation.

Dihybrid gives Independent Assortment.

EXAM TRAP -

Segregation aur Independent Assortment ko interchange mat karna.

3. Monohybrid Cross — 3:1 / 1:2:1

- P generation = TT Tall x tt Dwarf
- F1 = all Tt Hybrid Tall
- F1 self = Tt x Tt

F2 Ratio	Result
Phenotype	3 : 1 = Tall : Dwarf
Genotype	1 : 2 : 1 = TT : Tt : tt

TRICK - “Mono = P3, G121”

Phenotype 3:1; Genotype 1:2:1.

EXAM TRAP -

3:1 phenotype hai; 1:2:1 genotype.

4. Dihybrid Cross — 9:3:3:1

- P = RRYy Round Yellow x rryy Wrinkled Green
- F1 = RrYy Round Yellow
- F1 gametes = RY, Ry, rY, ry
- F2 phenotype ratio = 9:3:3:1

TRICK - “Di = 9-3-3-1”

Round Yellow : Round Green : Wrinkled Yellow : Wrinkled Green.

EXAM TRAP - “BbEe x BbEe > BBEE = 1/16”

Source ka direct probability trap.

5. Inheritance Scientists — S-B-M

Scientist	Contribution
Sutton & Boveri	Chromosomal theory of inheritance
T.H. Morgan	Drosophila experimental verification + gene theory
T.H. Morgan	Genes linear on chromosomes like beads

TRICK - “S-B Theory, Morgan Proof + Line”

Sutton-Boveri = theory.

Morgan = Drosophila verification + linear gene arrangement.

EXAM TRAP -

Chromosomal theory = Sutton & Boveri; Morgan = experimental verification/gene theory.

6. DNA Scientist Timeline — M-HW-WC

Scientist	Year	Fact
Friedrich Miescher	1869	Nucleic acid discovery
Hershey & Chase	1952	DNA as genetic material
Watson & Crick	1953	Double-helix model

TRICK - “69 Miescher - 52 Hershey/Chase - 53 Watson/Crick”

Discovery, proof, model — three different facts.

EXAM TRAP -

Watson-Crick = model; Hershey-Chase = genetic material proof.

7. Nucleoside vs Nucleotide — +P Rule

- Nucleoside = nitrogenous base + deoxyribose sugar
- Nucleotide = nucleoside + phosphate

TRICK - “Nucleotide = Nucleoside + P”

P = phosphate.

8. DNA Bases — AG Purines / TC Pyrimidines

Group	Bases
Purines	Adenine (A) + Guanine (G)
Pyrimidines	Thymine (T) + Cytosine (C)
Pairing	A-T and G-C

TRICK - "PURE As Gold = A-G"

Purines = A + G.

Pyrimidines = T + C.

TRICK - "AT / GC = DNA Couples"

Source base-pairing rule.

9. DNA vs RNA — DDT vs SRU

Feature	DNA	RNA
Strands	Double	Usually Single
Sugar	Deoxyribose	Ribose
Special base	Thymine	Uracil replaces Thymine
RNA types	—	rRNA ~80%, tRNA 10-15%, mRNA 3-5%

TRICK - "DNA = Double-Deoxy-Thymine; RNA = Single-Ribose-Uracil"

D-D-T vs S-R-U.

TRICK - "rRNA 80 > tRNA 10-15 > mRNA 3-5"

Source abundance order.

EXAM TRAP -

RNA has Uracil, not Thymine.

10. Gene Expression — R-T-T

• Central flow in source: DNA > Replication > DNA > Transcription > mRNA > Translation > Protein

TRICK - "Replicate DNA, Transcribe RNA, Translate Protein"

R-T-T sequence.

11. Honeybee Ploidy — Male n, Female 2n

TRICK - "Male = Mini n; Female = Full 2n"

Honeybee male haploid; female diploid.

EXAM TRAP -

Male honeybee = haploid, female = diploid.

12. Genetic Diseases — Sex-linked Pair

Disease	Cause / Type	Key Fact
Haemophilia	X-linked recessive	A = Factor VIII; B = Factor IX
Colour blindness	X-linked recessive	Red-green distinction affected

TRICK - "H-A = 8; H-B = 9"

Haemophilia-A Factor VIII, Haemophilia-B Factor IX.

TRICK - "X-linked = Males More Expressed"

Source reason: males have only one X chromosome.

13. Sickle Cell — 11 / 6 / Valine

- Sickle-cell anaemia = beta-globin gene mutation
- Gene on chromosome 11
- Valine replaces glutamic acid at 6th position
- RBC becomes sickle-shaped and carries less O₂

TRICK - "Sickle = 11-6-Val"

Chromosome 11, position 6, Valine.

EXAM TRAP -

Sickle cell chromosome = 11; substitution at 6th position.

14. Down - Turner - Klinefelter — 47 / 45 / 47

Syndrome	Chromosome Pattern	Sex / Cue
Down	Trisomy 21; total 47	Autosomal trisomy
Turner	44 + X0; total 45	Female
Klinefelter	44 + XXY; total 47	Male

TRICK - "D-T-K = 47-45-47"

Down 47, Turner 45, Klinefelter 47.

TRICK - "Turner = X0 Female; Kline = XXY Male"

Sex-chromosome patterns ko name ke saath lock karo.

EXAM TRAP -

Down = Trisomy 21; Turner = 45 female; Klinefelter = 47 male.

15. Evolution — Descent with Modification

- Biological evolution = complex organisms ka simple/primitive life forms se development
- Also called Descent with modification

TRICK - "Evolution = Descent + Modification"

Source ka direct alternate phrase.

16. Fossil Dating — C-U-K

- Fossils = dead organisms ke remains preserved in primitive rocks of Earth crust
- Age determination methods = C14 dating, U-Pb dating, K-Ar dating

TRICK - "C-U-K = Carbon - Uranium/Lead - Potassium/Argon"

Three dating methods source list se.

17. Homologous vs Analogous vs Vestigial

Type	Structure/Origin	Function	Example
Homologous	Similar	Different	Forelimbs of vertebrates
Analogous	Different	Similar	Wings of insects and birds
Vestigial	Functional in ancestors	Reduced/non-functional now	Appendix, wisdom teeth, coccyx

TRICK - “Homo = Same Structure, Different Job”

Homologous > divergent evolution.

TRICK - “Analog = Different Structure, Same Job”

Analogous > convergent evolution.

EXAM TRAP -

Homologous divergent; Analogous convergent.

18. Connecting Links — V-E-A

Organism	Connects
Virus	Living - Non-living
Euglena	Animal - Plant
Archaeopteryx	Reptile - Bird

TRICK - “V-E-A = Virus Life/Nonlife; Euglena Animal/Plant; Archaeopteryx Reptile/Bird”

Three direct connecting-link pairs.

19. Evolution Theories — L-D-V

Scientist	Book / Experiment	Core Idea
Lamarck	Philosophie Zoologique	Acquired characters; use & disuse
Darwin	On the Origin of Species, 1859	Natural selection; struggle; survival of fittest
Hugo de Vries	Evening Primrose	Mutation theory; sudden inherited changes

TRICK - “L-D-V = Lamarck Acquired, Darwin Selection, de Vries Mutation”

Theory-scientist pairing.

TRICK - “Lamarck = Use/Disuse; Darwin = Natural Selection; Vries = Sudden Mutation”

Core-idea chain.

EXAM TRAP -

Darwin 1859 Origin; Lamarck Philosophie Zoologique; de Vries Evening Primrose.

20. Final Exam Traps — 15 Red Flags

- Segregation = Purity of Gametes; Independent Assortment = dihybrid
- Monohybrid phenotype 3:1; genotype 1:2:1
- Dihybrid phenotype 9:3:3:1; BBEE probability 1/16 in source cross
- Sutton-Boveri theory; Morgan Drosophila verification/gene theory
- Miescher 1869; Hershey-Chase 1952; Watson-Crick 1953
- RNA = Uracil, not Thymine
- Honeybee male n; female 2n
- Haemophilia-A VIII; Haemophilia-B IX

- Sickle cell = chromosome 11, Valine for glutamic acid at 6th
- Down = Trisomy 21, 47
- Turner = 44+X0, 45, female
- Klinefelter = 44+XXY, 47, male
- Homologous same structure/different function = divergent
- Analogous different structure/same function = convergent
- Archaeopteryx reptile-bird; Euglena animal-plant; Virus living-non-living

EXAM TRAP - "Laws-Ratios-Scientists-RNA-Ploidy-Disease-Evolution"

Chapter ke highest-confusion areas isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"M-P-7 = Mendel-Pisum-7 pairs; dominant T-R-Y-P vs recessive D-W-G-W → laws D-S-I; Segregation=Purity, Mono laws1+2, Di law3 → Monohybrid P3/G121; Dihybrid 9-3-3-1; BbEe BBEE=1/16 → Sutton-Boveri theory, Morgan Drosophila proof+linear genes → DNA timeline Miescher1869 > Hershey-Chase1952 > Watson-Crick1953; nucleoside base+sugar, nucleotide +phosphate; Purines AG, Pyrimidines TC, pairs AT/GC → DNA D-D-T vs RNA S-R-U; rRNA80 > tRNA10-15 > mRNA3-5; Replicate-Transcribe-Translate → Honeybee male n/female2n → Haemophilia A8/B9 + colour blindness X-linked; Sickle 11-6-Val; D-T-K = Down47/21, Turner45-X0-female, Kline47-XXY-male → Evolution = descent with modification; fossils C-U-K dating → Homologous same/different divergent; Analogous different/same convergent; vestigial appendix-wisdom-coccyx → links V-E-A = Virus life/nonlife, Euglena animal/plant, Archaeopteryx reptile/bird → theories L-D-V = Lamarck acquired/use-disuse, Darwin natural selection/Origin1859, de Vries mutation/Evening Primrose."

TOP 5: Mono 3:1 & 1:2:1 vs Di 9:3:3:1 | Miescher-Hershey/Chase-Watson/Crick | RNA U not T | D-T-K 47-45-47 | Homologous divergent vs Analogous convergent.

HUMAN DISEASE — MEMORY TRICK SHEET

General Science · Biology · Chapter 10 · Disease Types + Transmission + Pathogens + Vectors + Diagnostic Tests

Ye full notes nahi hai — source chapter ke disease-type distinctions, transmission routes, viral/bacterial/protozoan/fungal/worm diseases, vectors, diagnostic tests aur high-confusion disease-pathogen pairs ko fast exam recall format mein compress kiya gaya hai.

1. Disease Types — C-A-I / Non-Infectious

Type	Source Meaning / Examples
Congenital	Genetic/developmental defects during embryonic development; cerebral palsy, haemophilia, colour blindness
Acquired - Acute	Short-term; cold, cholera
Acquired - Chronic	Long-lasting; TB, elephantiasis, arthritis
Infectious	Spread through contact/carrier/medium; pathogens include virus, bacteria, protozoa, fungi, worms

TRICK - “C-A-C-I = Congenital - Acute - Chronic - Infectious”

Acute = short; Chronic = long.

Infectious = pathogen + transmission route.

2. Transmission — 5F Rule

- Air droplets from coughing/sneezing
- Contaminated food and water
- Contaminated blood/body fluids
- Disease vectors such as flies and mosquitoes
- Indirect contact through contaminated objects

TRICK - “5F = Flies - Fingers - Fomites - Food - Fluid”

Source ka ready-made transmission recall rule.

EXAM TRAP -

Fomites = contaminated objects ke through indirect contact, source framing ke according.

3. Non-Infectious Disease Traps — As-Cd-F

Category	Examples / Pair
Allergies	Hay-fever, Asthma
Nutritional deficiency	Marasmus-protein; Goitre-iodine; Pellagra-B3; Dental cavities-fluoride
Degenerative	Heart attack, Arthritis, Diabetes, Glaucoma, Cataract
Mental	Stress, Depression, Epilepsy, Anorexia nervosa
Pollution-related	Arsenic-Black foot; Cadmium-Itai-itai; Fluoride-Knock-Knee
Cancer	Uncontrolled cell growth; Leukemia, Carcinoma, Sarcoma

TRICK - “As Black - Cd Itai - F Knock”

Arsenic = Black foot disease.

Cadmium = Itai-itai.

Fluoride water = Knock-Knee syndrome.

EXAM TRAP -

Arsenic/Cadmium/Fluoride disease pairs ko interchange mat karna.

4. Viral Diseases — Air, Aedes, Food, Bite, Blood

Disease	Pathogen	Transmission
Small Pox	Variola virus	Air borne
Chicken Pox	Varicella Zoster	Air borne
Measles / Rubella	Rubellavirus	Air borne - as stated in source
Influenza	Orthomyxovirus	Air borne
Swine Flu	Influenza A subtype H1N1	Air borne
Yellow Fever	Flavi virus	Aedes aegypti bite
Dengue	DENV	Aedes aegypti bite
Chikungunya	CHIKV	Aedes aegypti bite
Poliomyelitis	Polio RNA virus type 1,2,3	Contaminated soil/water/food
Rabies / Hydrophobia	Lyssa virus type-1	Infected animal bite
AIDS / Slim disease	Retrovirus HIV	Unsafe sex, blood/needles/razor, mother-to-foetus

TRICK - “Aedes = D-C-Y”

Dengue + Chikungunya + Yellow Fever.

TRICK - “Swine = H1N1; Rabies = Hydrophobia; AIDS = HIV + ELISA”

Three viral direct-MCQ pairs.

EXAM TRAP -

Polio is stated as eradicated in India in the source notes.

5. Viral Extra Facts — STD + COVID + Jaundice

- HIV-AIDS, HPV and Hepatitis-B = Sexually Transmitted Diseases in source notes
- COVID-19 = zoonotic disease caused by SARS-CoV-2; source notes mention animal origin
- Jaundice = yellow skin; symptom associated with hepatitis/liver inflammation

TRICK - “STD Trio = HIV - HPV - Hep-B”

COVID = SARS-CoV-2.

Jaundice = symptom, source connects it with hepatitis/liver inflammation.

6. Bacterial Diseases — Air vs Food/Water vs Other

Disease	Pathogen	Transmission
TB	Mycobacterium tuberculosis	Air borne
Diphtheria	Corynebacterium diphtheriae	Air borne
Whooping cough	Bordetella pertussis	Air borne
Pneumonia	Streptococcus pneumoniae	Air borne
Cholera	Vibrio cholerae	Contaminated food/water
Typhoid	Salmonella typhi	Contaminated food/water; Widal test
Diarrhoea	Salmonella sp. + E. coli; source also notes Rotavirus	Contaminated food/water
Plague / Black Death	Yersinia pestis	Rat fleas
Tetanus	Clostridium tetani	Wound contamination
Leprosy / Hansen disease	Mycobacterium leprae	Direct contact

TRICK - “Typhoid WIDAL; AIDS ELISA”

Diagnostic-test pair sabse important.

TRICK - “TB + Leprosy = Mycobacterium”

tuberculosis vs leprae species.

TRICK - “Plague Rat Flea; Tetanus Lock Jaw; Leprosy Hansen”

Plague = Black Death.

Tetanus = Lock Jaw / 8th Day disease.

EXAM TRAP -

Plague = rat flea, mosquito nahi.

7. Bacterial Extra — Trachoma + Gastritis

- Trachoma = eye disease that may lead to blindness; Chlamydia trachomatis
- Gastritis in stomach = Helicobacter pylori

TRICK - “Trachoma-Eye-Chlamydia; Gastritis-Stomach-H. pylori”

Location + pathogen pair.

8. Protozoan Diseases — Water + Vector Chain

Disease	Pathogen	Transmission
Diarrhoea	Giardia intestinalis	Contaminated water/food
Amoebiasis	Entamoeba histolytica	Contaminated water/food
Pyorrhoea	Entamoeba gingivalis	Saliva / Kiss
Kala-azar	Leishmania donovani	Sand fly
Malaria	Plasmodium sp.	Female Anopheles
African Sleeping Sickness	Trypanosoma gambiense / T. brucei	Tse-tse fly

TRICK - “K-M-S = Kala Sand, Malaria Anopheles, Sleeping Tse-tse”

Vector trio: Sand fly > Kala-azar; Female Anopheles > Malaria; Tse-tse > Sleeping sickness.

EXAM TRAP -

Malaria = FEMALE Anopheles; not male, not Aedes.

9. Worm + Fungal Diseases

Group / Disease	Pathogen	Transmission
Ascariasis - Worm	Ascaris lumbricoides	Contaminated fruits/vegetables
Elephantiasis / Filariasis - Worm	Wuchereria bancrofti	Culex mosquito
Ringworm - Fungal	Microsporum + Trichophyton	Direct/indirect contact
Candidiasis - Fungal	Candida	Direct/indirect contact

TRICK - “Ascaris = Ascariasis; Wuchereria = Elephantiasis”

Worm disease pair.

EXAM TRAP - “Ringworm = FUNGUS, Not Worm”

Microsporum + Trichophyton.

10. Mosquito Master — ADCy / AM / CE

Mosquito	Disease(s)	Pathogen
Aedes aegypti	Dengue + Chikungunya + Yellow Fever	DENV, CHIKV, Flavi virus
Female Anopheles	Malaria	Plasmodium sp.
Culex	Elephantiasis / Filariasis	Wuchereria bancrofti

TRICK - “ADCy = Aedes-Dengue-Chikungunya-Yellow”

Aedes ka three-disease pack.

TRICK - “AM = Anopheles-Malaria; CE = Culex-Elephantiasis”

Female Anopheles = Malaria.

Culex = Elephantiasis/Filariasis.

EXAM TRAP -

Aedes, Anopheles aur Culex ke diseases ko mix mat karna.

11. Other Vectors — S-T-R-D

Vector	Disease	Pathogen
Sand fly	Kala-azar	Leishmania donovani
Tse-tse fly	African Sleeping Sickness	Trypanosoma gambiense / T. brucei
Rat flea	Plague / Black Death	Yersinia pestis
Dog/Wolf/Monkey bite	Rabies / Hydrophobia	Lyssa virus type-1

TRICK - “S-T-R-D = Sand-Kala, Tse-Sleep, Rat-Plague, Dog-Rabies”

Vector-bite chain ko sequence mein yaad karo.

12. Diagnostic Tests — E-W

Test	Disease
ELISA	AIDS / HIV
Widal Test	Typhoid

TRICK - “E = HIV; W = Typhoid”

ELISA-AIDS/HIV; Widal-Typhoid.

EXAM TRAP -

ELISA aur Widal ko interchange mat karna.

13. Pathogen-Wise Master Buckets

Group	Disease Chain from Source
Viral	Small/Chicken Pox; Measles/Rubella; Influenza; Swine H1N1; Dengue; Chikungunya; Yellow Fever; Polio; Rabies; AIDS
Bacterial	TB; Diphtheria; Whooping cough; Pneumonia; Cholera; Typhoid; Plague; Tetanus; Leprosy; Trachoma; Gastritis
Protozoan	Malaria; Kala-azar; Amoebiasis; Pyorrhoea; African Sleeping Sickness
Fungal + Worm	Ringworm; Candidiasis; Ascariasis; Elephantiasis

TRICK - “V-B-P-FW”

Viral > Bacterial > Protozoan > Fungal/Worm buckets mein disease recall karo.

14. Final Exam Traps — 12 Red Flags

- 5F = Flies, Fingers, Fomites, Food, Fluid
- Arsenic-Black foot; Cadmium-Itai-itai; Fluoride-Knock-Knee
- Aedes = Dengue + Chikungunya + Yellow Fever
- Female Anopheles = Malaria

- Culex = Elephantiasis / Filariasis
- Sand fly = Kala-azar
- Tse-tse fly = African Sleeping Sickness
- Rat flea = Plague / Black Death
- ELISA = AIDS/HIV; Widal = Typhoid
- Ringworm = fungal disease, not worm disease
- Rabies = Hydrophobia; infected animal bite
- Tetanus = Lock Jaw / 8th Day disease; Leprosy = Hansen disease

EXAM TRAP - "5F - Pollution - 3 Mosquitoes - 3 Flies/Flea - Tests - Names"

Chapter ke most-asked traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Disease types C-A-C-I: congenital, acquired acute-short, chronic-long, infectious-pathogen → transmission 5F = Flies-Fingers-Fomites-Food-Fluid → noninfectious pollution: As-Black foot, Cd-Itai-itai, Fluoride-Knock-Knee; cancer uncontrolled growth → viral: Smallpox-Variola, Chickenpox-Varicella, Flu-Orthomyxo, Swine-H1N1, Aedes D-C-Y, Polio contaminated food/water, Rabies Hydrophobia-Lyssa, AIDS HIV-ELISA; STD HIV-HPV-HepB; COVID SARS-CoV-2; Jaundice hepatitis/liver symptom → bacterial: TB Mycobacterium tuberculosis, Typhoid Salmonella-Widal, Cholera Vibrio, Plague Yersinia-rat flea, Tetanus Clostridium-Lock Jaw, Leprosy M.leprae-Hansen; Trachoma Chlamydia eye, Gastritis H.pylori → protozoan: Giardia diarrhoea, Entamoeba amoebiasis/gingivalis pyorrhoea, Kala-azar Sand-Leishmania, Malaria Female Anopheles-Plasmodium, Sleeping Tse-tse-Trypanosoma → worms: Ascaris ascariasis, Wuchereria filariasis-Culex; fungi: Ringworm Microsporum+Trichophyton, Candidiasis Candida → mosquito code ADCy / AM / CE; other vectors Sand-Kala, Tse-Sleep, Rat-Plague, Dog-Rabies → tests ELISA-HIV, Widal-Typhoid."

TOP 5: Aedes D-C-Y | Female Anopheles Malaria | Culex Filariasis | ELISA HIV vs Widal Typhoid | Ringworm = fungus, not worm.